

Role Uncertainty and Punishment Severity in Repeated Cheap Talk: Theory and Experiment*

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We investigate the efficacy of the shadow of the future in sustaining information transmission under partially aligned interests by manipulating two institutional features: punishment severity and role uncertainty. While theory suggests that these features potentially facilitate fully revealing equilibria relative to a baseline of fixed roles and mild punishment, our experiment documents an asymmetric effect on sender and receiver behavior. Harsh punishment with fixed roles disciplines senders but fails to elicit trust from receivers. Conversely, mild punishment with role uncertainty fosters trust rather than honesty, supporting an outcome where senders obtain their preferred action without needing to be truthful. The combination of both features asymmetrically improves subjects' honesty as senders but discourages their trust as receivers. Our findings imply that mechanisms aiming for truthful and efficient communication must carefully tailor incentives for both roles, rather than relying solely on disciplining the informed sender.

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Strategic information transmission is central to economic interaction, yet often hindered by misaligned incentives between informed and uninformed parties. While static theory predicts communication breakdown (Crawford and Sobel, 1982), many real-world interactions are ongoing. This repetition offers a potential remedy: the “shadow of the future” allows for fully revealing equilibria supported by conditional strategies. However, theoretical possibility does not ensure empirical success; coordinating on such contingent strategies proves difficult in practice (Wilson and Vespa, 2020). Accordingly, understanding which environmental features facilitate or hinder information transmission in repeated interactions is crucial.

This paper examines two realistic features. The first feature, *role uncertainty*, captures the dynamic nature of long-term relationships: individuals may alternate between being the informed sender and the decision-maker. The second feature, *punishment severity*,

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reflects the cost of communication breakdowns, where deviations may trigger a permanent reversion to a less favorable outcome. These features arise naturally in professional networks where parties repeatedly exchange assessments and alternate between roles with partially aligned interests.

One leading example that motivates our setting involves investment syndication. In many financial environments, counterparties repeatedly share access to deal flow. In any given period, the originator holds private information about the quality of an investment opportunity, while the investor must decide whether to participate. The originator typically benefits from the counterparty’s participation through origination fees or capital leverage, irrespective of the opportunity’s true quality. In contrast, the counterparty gains only when the opportunity is genuinely valuable. Crucially, firms frequently alternate between originating deals and investing in others’ projects, creating a natural structure of role uncertainty.

This structure extends beyond finance to the academic referral system, particularly in graduate school admissions. Professors frequently exchange evaluations of each other’s undergraduate students. The recommender possesses private information about an applicant’s potential and communicates it through the reference letter. While a recommender may have an incentive to secure an admission offer to boost their student’s prospects regardless of true aptitude, the admission committee benefits only from admitting a candidate who is a high-quality match. Because such interactions recur across admission cycles and professors alternate between writing letters and serving on admission committees, these long-term relationships incorporate role uncertainty.

A distinct but related dynamic appears in the “revolving door” phenomenon between regulatory agencies and the private sector especially in law and financial industries.¹ Firms possess private information regarding their compliance levels or asset quality and communicate this to regulators who must rely on these disclosures to make enforcement or approval decisions. However, regulators often transition into industry jobs later in their careers, while industry professionals may enter government service (for detailed discussions, see deHaan et al. (2015); Brezis (2017); Wirsching (2018); Luechinger and Moser (2023), among others). This creates a dynamic of role uncertainty where today’s decision-maker may be tomorrow’s reporting agent. Furthermore, the severity of punishment for deceptive reporting or compliance failures varies significantly across jurisdictions ranging from mild fines to permanent industry bans. These institutional features, the probability of role switching and the severity of sanctions, define the incentives for truth-telling and trust within the regulatory relationship.

Across these diverse settings, the strategic information transmission hinges on two structural features: the likelihood of role reversal and the cost of trust breakdown. This heterogeneity generates distinct incentives for truthful communication, which we isolate and analyze in our experimental design.

We formalize these two features in an indefinitely repeated cheap talk environment with a fixed pair of players, A and B. The underlying stage game consists of two states, two messages, and three actions. In each period, one player acts as Sender and the other

¹ We thank Matthew Kovach for bringing up this example.

as Receiver. Sender privately observes the state and communicates it to Receiver, who then takes an action. Interests are partially aligned: Sender always favors one fixed action, while Receiver prefers matching the action with the state. In the one-shot stage game, this conflict of interest leads to a Nash outcome in which Receiver chooses the intermediate action, yielding payoffs of $(1 - b, b)$ to Sender and Receiver, respectively.

In a grim-trigger strategy with the stage-Nash outcome as punishment, the parameter $b \in [0.5, 1]$ governs the payoff distribution in the punishment phase. Specifically, a higher b implies a lower payoff for Sender and a higher payoff for Receiver, thereby indexing the severity of punishment on Sender.² We incorporate role uncertainty by letting $\gamma \in [0.5, 1]$ denote the probability that player A acts as Sender and player B acts as Receiver in any given period.³ Together, γ and b capture the key institutional features described in our motivating examples.

We study how these two forces affect the sustainability of informative communication, both individually and jointly. We focus on two truth-telling equilibria. In the *Simple Truth-Telling Equilibrium* (STE), Sender always reports truthfully, and Receiver “simply” follows the message, extracting the entire information surplus.⁴ In the *Information Rent Equilibrium* (IE), Sender again truthfully reports, but Receiver conditions her response on whether interests are aligned. She follows Sender’s message when incentives are aligned and plays the stage-Nash action when interests conflict, thereby granting Sender an “information rent” in the spirit of Wilson and Vespa (2020). This rent strengthens Sender’s incentive to be honest but requires Receiver to coordinate on a contingent sacrifice of immediate surplus. Both equilibria rely on grim-trigger strategies, and their feasibility depends on whether the excess of the continuation value of future cooperation over that of the punishment phase outweighs the immediate gain from deviation.

Our comparative static analysis focuses primarily on STE, indexed by the minimal required time discount factor to enable its feasibility. Among fully revealing equilibria, it serves as a natural benchmark due to its strategic simplicity. In our baseline case of fixed roles ($\gamma = 1$) and mild punishment ($b = 0.5$), STE is only sustainable without time discounting. Raising the cost of deviation for Sender (increasing b) theoretically improves sustainability. Introducing role uncertainty ($\gamma < 1$) also facilitates cooperation, as players alternate roles and share the surplus over time, reducing the incentive to misreport.

However, the interaction of role uncertainty and strong punishment creates opposing forces on the current Sender’s incentives, making STE’s comparative statics more nuanced. Stronger punishment increases the cost of lying when the player expects to be Sender, but simultaneously lowers the cost of lying when the player anticipates to be Receiver. The net effect of strong punishment depends on the degree of role uncertainty.

With symmetric role uncertainty ($\gamma = 0.5$), the opposing forces offset each other. With asymmetric roles ($\gamma \in (0.5, 1)$), the effect depends on the binding incentive constraint that determines STE’s minimal required time discount factor. Specifically, with asymmetric

²Throughout the paper, “stronger punishment” is with respect to Sender.

³Symmetric relationships, such as collaborations among peers, correspond to $\gamma = 0.5$, whereas more hierarchical environments often exhibit $\gamma > 0.5$.

⁴Receiver in STE does not have incentives to deviate as she is already taking the myopically optimal action.

role uncertainty, it is necessary to regulate the incentives to lie for both players when they act as current Sender in the misaligned state. When punishment is moderate ($b \in (0.5, 0.75)$), the constraint for the player who is more likely to be Sender binds; thus, increasing b relaxes the constraint and aids STE. Conversely, when punishment is already severe ($b \in [0.75, 1)$), the constraint for the player who is more likely to be Receiver binds, as her payoff in the punishment phase is actually high. In this region, increases in b further raise the value of the punishment phase for that player, thereby tightening their incentive constraint and potentially reducing sustainability of STE.

The comparative statics regarding IE are distinct. Introducing role uncertainty or stronger punishment individually does not affect its sustainability. When $b = 0.5$, Sender and Receiver face symmetric incentives, rendering role uncertainty irrelevant. Under fixed roles, the effects of changes in b on punishment severity and the size of the information rent offset each other. However, the interaction between role uncertainty and punishment becomes detrimental: when $b > 0.5$, Sender's incentive to deviate in misaligned states increases as their likelihood of being Sender decreases; when roles are random, Sender's incentive to deviate grows as b increases.

Furthermore, it is crucial to distinguish between the *theoretical* sustainability of IE and its *behavioral* viability. While IE is often theoretically easier to sustain than STE due to Sender's relaxed incentive constraint by the information rent, it is behaviorally more demanding. Coordinating on IE is empirically difficult because it introduces a *two-sided temptation*: Sender is tempted to lie, and Receiver is tempted to withhold the information rent by playing her myopic best response. It requires Receiver to willingly forgo short-term gains, demanding a level of strategic sophistication and mutual understanding that is hard to achieve. Indeed, as demonstrated by Wilson and Vespa (2020), subjects struggle to coordinate on such strategies without explicit coordination devices. In contrast, STE does not require such coordination and is more attainable. Therefore, we center our experimental analysis on whether the environmental features facilitate the strategically simpler STE.

We experimentally compare six treatments that vary along two dimensions: role uncertainty and punishment severity. Our **Baseline** treatment features fixed roles and mild punishment. Three treatments vary only one factor: Unequal-chance **Random** roles with mild punishment (**UR**), Equal-chance **Random** roles with mild punishment (**ER**), and **Harsh** punishment with fixed roles (**H**). Two treatments vary both factors: **Harsh** punishment with Unequal-chance **Random** roles (**HUR**) and **Harsh** punishment with Equal-chance **Random** roles (**HER**).

Based on our theoretical framework about whether STE can be supported in certain environment, we expect information transmission to be more efficient in H, ER, and HER than in the Baseline. The efficiency is defined as the frequency of Receiver's action matching the state when interest are aligned. We expect no significant difference in UR and a lower efficiency in HUR compared to the Baseline. Moreover, we expect similar outcomes in ER and HER, since punishment severity has no additional effect on the sustainability of STE under symmetric role uncertainty. We report three main findings from our data analysis.

First, we find that increasing punishment severity introduces a behavioral friction we term the “safety trap”: when the intermediate action offers a secure, non-contingent payoff, raising that payoff leads Receivers to default to this action and to avoid other message-state-contingent choices, even when messages become more reliable. The evidence comes from Treatment H. Harsh punishment disciplines Senders to be truthful, but because the same parameter b also determines Receiver’s sure payoff from the intermediate action, harsher punishment simultaneously raises Receiver’s opportunity cost of trust. Fearing coordination failure, some Receivers therefore retreat to the intermediate action despite Sender’s increased honesty. Thus, while harsh punishment resolves Sender’s incentive problem, it tightens Receiver’s trust trade-off, resulting in no net gain in efficiency.

Second, we find that mild punishment combined with random roles (ER, UR) fosters an efficient but uninformative outcome consistent with the *Blind Trust Equilibrium* (BTE). In this equilibrium, Receivers trust even imperfect signals, and Senders need not be truthful. Unlike truth-telling equilibria which require strict adherence to honesty, BTE relies on a mechanism of reciprocity: players take turns to tolerate uninformative messages and enjoy preferred action as Sender. This outcome demonstrates that in environments with role uncertainty and low stakes, subjects can bypass the cognitive complexity of contingent truth-telling strategies and achieve high efficiency through mutual forbearance.

Third, we examine the interaction between harsh punishment and role uncertainty (HUR, HER). Contrary to the theoretical prediction that HER can facilitate cooperation, we do not find evidence that it improves efficiency relative to the Baseline; if anything, efficiency declines. A key pattern is a larger decrease in Receiver trust than under harsh punishment with fixed roles, which is consistent with an amplified “safety trap.” We observe a breakdown in strategic consistency where subjects decouple their incentives across roles. Subjects tend to comply with truth-telling as Senders to avoid harsh penalties, yet default to the intermediate action as Receivers because its sure payoff is high. One intuitive reason is that choosing the intermediate action can feel less like imposing a persistent payoff disadvantage on the counterpart: any ex post inequality within a given round may be offset ex ante by future role realizations, which can weaken other-regarding restraint and facilitate a retreat to safety.

Consistent with the “safety trap” driven by the larger b , a dynamic analysis of within-game play further reveals that the retreat to the intermediate action is not primarily a punitive response to a Sender’s deception. While such conditional punishment does occur, in the majority of cases, distrust manifests either as an immediate, myopic reaction from the very first round, or as an unprovoked abandonment of established cooperation. These dynamics highlight that the intermediate action serves more as a default absorbing state driven by strategic uncertainty than as a conditional punishment device to enforce truth-telling equilibria.

A critical insight from these behavioral patterns extends beyond the theoretical focus on Sender’s incentives. While the experimental design was motivated by the interplay between punishment and role uncertainty in disciplining the informed party, our data re-

veal a fundamental asymmetry: the primary obstacle to efficiency is often not Sender’s temptation to lie, but Receiver’s reluctance to trust. We find that mechanisms designed to strictly bind Sender (e.g., harsh punishment) can backfire if it leads to a more conservative trust. Specifically, the implicit assumption in standard theory that Receiver’s cooperation is automatic once Sender is disciplined needs not hold when the strategic risk of trust becomes too high. Conversely, the introduction of role uncertainty (under mild punishment) improves efficiency by rendering this assumption obsolete, as BTE paves the way for a distinct form of cooperation without the necessity of honesty.

These findings underscore a fundamental distinction between efficiency and truthfulness. While truth-telling equilibria are sufficient for maximal efficiency, they are not necessary; high efficiency can also arise from “blind trust” despite imperfect information. The reason is that to achieve higher efficiency in the long run, role uncertainty opens the gate for various reciprocal turn-taking mechanisms, with or without the need for truth-telling. While STE requires turn-taking to be honest, BTE requires turn-taking to turn a blind eye. This decoupling suggests that institutional design must be goal-dependent. If the objective is to maximize social surplus, environments that foster tolerance and blind trust may be preferable. However, if the objective is to elicit honest reporting and efficient communication as in auditing or regulatory contexts, distinct incentives are required to prevent the “safety trap” where honesty is met with distrust.

In the remainder of the paper, Section I reviews the related literature. Section II presents the theoretical model and hypotheses. Section III describes the experimental design. Sections IV and V present the aggregate empirical findings and the subject-supergame level analysis. Section VI concludes.

I. Literature

This paper contributes to the literature on effective communication in cheap talk settings. Building on the foundational result of limited information transmission in static interactions between a privately informed sender and a decision-making receiver (Crawford and Sobel, 1982), researchers have explored various conditions that promote communication in static games (see Blume, Lai and Lim (2020) for a review) and examined the potential of repeated interactions to sustain informative outcomes (Goloso et al., 2014; Wilson and Vespa, 2020; Kolotilin and Li, 2021; Kuvalekar, Lipnowski and Ramos, 2022; Best and Quigley, 2024), among others.⁵

Our study contributes to this line of research by examining how realistic institutional features affect information transmission in repeated cheap talk games. Specifically, we focus on punishment severity and role uncertainty that are both theoretically relevant and empirically observable.

⁵Without observable history, pure relational incentives are generally insufficient to restore full commitment in repeated games (Kuvalekar, Lipnowski and Ramos, 2022). When history is available, fully revealing equilibria can be supported (Goloso et al., 2014). The quality and structure of historical information also influence truth-telling behavior (Best and Quigley, 2024). Additionally, transfers can facilitate communication by acting as signaling devices, as well as mechanisms to discipline Receiver (Kolotilin and Li, 2021) or to incentivize Sender towards full revelation (Wilson and Vespa, 2020).

Role uncertainty has been explored in related experimental settings such as dictator games and trust games. Existing experiments have produced mixed findings on the effect of potential role switching on altruism (Classen, 2005; Iriberri and Rey-Biel, 2011),⁶ as well as on reciprocity (Ben-Ner et al., 2004; Herne, Lappalainen and Kestilä-Kekkonen, 2013; Burks, Carpenter and Verhoogen, 2003; Johnson and Mislin, 2011). These studies suggest that role uncertainty can reduce altruistic behavior: when the dictator role is not permanent, the current dictator may feel less responsible for the recipient's payoff, making self-interested behavior more normatively acceptable. At the same time, the possibility of switching roles can promote reciprocal thinking, which in turn encourages cooperation. Similarly, our theoretical analysis shows that role uncertainty influences the *ex ante* distribution of private information, punishment power, and information surplus. In addition, our experiment finds that role uncertainty amplifies the “safety trap” induced by severe punishment (high value of the intermediate action): when the intermediate action provides a highly attractive sure payoff, role switching can reduce the salience of within-round payoff asymmetries (because expected payoffs are *ex ante* balanced by exogenous role assignment), potentially strengthening the pull of the trap by making self-interested behavior more justifiable.

Finally, our paper is most closely related to and directly builds upon the seminal work of Wilson and Vespa (2020). They highlight the fundamental coordination failures in repeated cheap talk games and show how specific institutional features, such as coordination devices and asymmetric punishment, can restore efficiency. We extend their framework by incorporating different levels of punishment severity and random role assignments. By doing so, we contribute to this literature by uncovering additional behavioral frictions and demonstrating that role uncertainty can decouple efficiency from honesty via BTE. Our work complements existing studies by showing that balancing the incentives and behavioral motives of *both* roles is crucial for institutional design in repeated communication.

II. Theory, Design and Hypotheses

The general repeated cheap talk game involves a fixed pair of players, A and B. The game unfolds over an indefinite number of rounds, with $\delta \in (0, 1)$ denoting the probability of continuation. Equivalently, the game can be viewed as having an infinite horizon with a common discount factor δ . In each round, the players engage in the following sequence of actions. First, Nature assigns roles: with probability $\gamma \in [0.5, 1]$, A is Sender and B is Receiver; with probability $1 - \gamma$, A is Receiver and B is Sender. Next, Nature draws a state $\theta \in \Theta$, observed only by Sender. Sender sends a message $m \in \mathcal{M}$ to Receiver, who then chooses an action $a \in \mathcal{A}$. Payoffs depend on the state and Receiver's action. Full histories of roles, states, messages, and actions in the previous rounds are available to both players.

Let t index the rounds. Players are risk-neutral. Player A's objective is to maximize

⁶Mesa-Vázquez, Rodríguez-Lara and Urbano (2021) further show that the effect of role uncertainty on altruism is sensitive to the framing of the game.

her discounted expected payoff:

$$U_A \equiv \sum_{t=1}^{\infty} \delta^{t-1} \{ \gamma \mathbb{E}_{\theta} [u^S(a_t(m_t(\theta)), \theta)] + (1 - \gamma) \mathbb{E}_{\theta} [u^R(a_t(m_t(\theta)), \theta)] \},$$

where $u^S(\cdot)$ and $u^R(\cdot)$ are the per-round payoffs when the player acts as Sender or Receiver, respectively.

Player B maximizes:

$$U_B \equiv \sum_{t=1}^{\infty} \delta^{t-1} \{ (1 - \gamma) \mathbb{E}_{\theta} [u^S(a_t(m_t(\theta)), \theta)] + \gamma \mathbb{E}_{\theta} [u^R(a_t(m_t(\theta)), \theta)] \}.$$

We simplify the environment to two states, two messages, and three actions.

A. The Stage Game

At the start of each round, roles are assigned and players play a static cheap talk game with the extensive form displayed in Figure 1. The state space is $\Theta = \{L, R\}$, message space $\mathcal{M} = \{“L”, “R”\}$, and action space $\mathcal{A} = \{L, M, R\}$. The prior is uniform: $\mathbb{P}(\theta = L) = \mathbb{P}(\theta = R) = 0.5$. Sender’s payoff is:

$$u^S(a) = \begin{cases} 1 & \text{if } a = L, \\ 1 - b & \text{if } a = M, \\ 0 & \text{if } a = R, \end{cases}$$

and Receiver’s payoff is:

$$u^R(\theta, a) = \begin{cases} 1 & \text{if } (\theta, a) \in \{(L, L), (R, R)\}, \\ b & \text{if } a = M, \\ 0 & \text{otherwise.} \end{cases}$$

Sender always prefers action L , while Receiver prefers matching the state *ex post* and *M ex ante* (given $b \geq 0.5$). In the static game, this misalignment leads to babbling as the unique equilibrium: Receiver ignores the message and plays M . While she may randomize when $b = 0.5$, a small positive increment ε to b can break the tie; thus, any $b > 0.5$ guarantees M is optimal. This *Babbling Equilibrium* (BE) yields an aggregate payoff of 1 in both states which is less than the efficient truth-telling outcome, whose aggregate payoff is 2 in state L and 1 in state R .

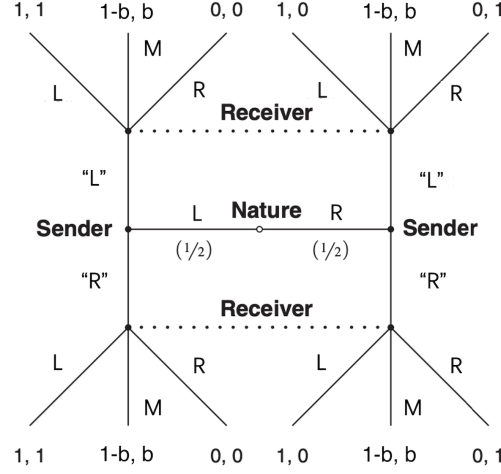


Figure 1. : The stage game.

B. The Simple Truth-Telling Equilibrium

In repeated play, more informative equilibria become feasible. We focus on STE.⁷ In STE, Sender tells the truth, and Receiver follows the message. Any deviation triggers a permanent reversion to BE. The equilibrium is “simple” in that Receiver always chooses her myopically optimal action given the message, extracting the full surplus without needing explicit reward-sharing. We assume symmetric strategies across players. That is, a player’s strategy is independent of their identity (Player A or B): both players adopt the exact same behavioral rule when acting as Sender, and the same behavioral rule when acting as Receiver, regardless of their ex-ante probability of being assigned either role.

On-path, neither player has an incentive to deviate when acting as Receiver. Sender, however, may benefit by misreporting in state *R*, gaining 1 if Receiver follows. Deviation triggers punishment (action *M*), yielding off-path payoffs of $(1 - b, b)$. Thus, larger b implies a harsher punishment for Sender.

STE-BASED COMPARATIVE STATICS. — We define $\hat{\delta}$ as the minimum discount factor required to support STE for a given (γ, b) . Smaller $\hat{\delta}$ indicates easier sustainment.

To derive $\hat{\delta}$, we examine profitable deviations by Senders in state *R*. The gain from deviation is always 1; the future cost C in each round for player A and B respectively depends on both b and γ . Figure 2 plots the smaller of the two possible values of C corresponding to the binding incentive constraint. The condition to prevent profitable deviation is $(\delta C)/(1 - \delta) \geq 1$, so the binding constraint determines $\hat{\delta}$.

Figure 3 shows the minimal required discount factor $\hat{\delta}$ to sustain STE across different combinations of punishment severity (b) and role uncertainty (γ). The horizontal axis

⁷A full analysis is provided in Appendix A.A1.

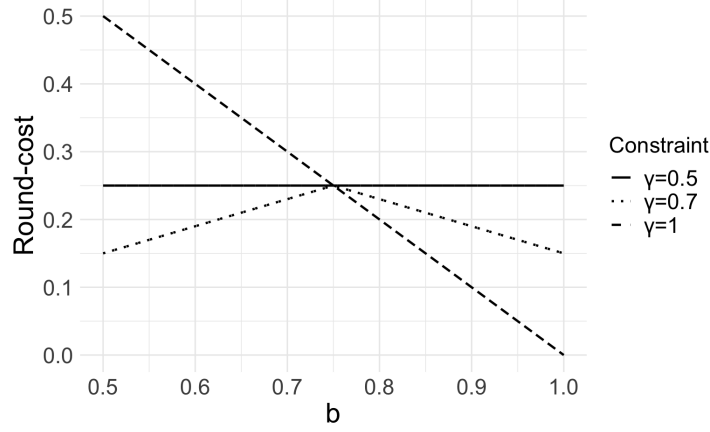


Figure 2. : Round-cost of deviation under different (γ, b) .

represents b , and the vertical axis represents $\hat{\delta}$. The point at the top-left corner of the figure—fixed roles ($\gamma = 1$) and mild punishment ($b = 0.5$)—serves as our baseline for comparison.

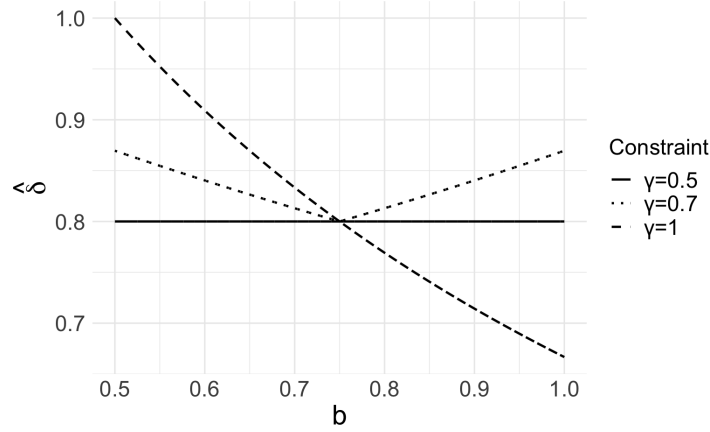


Figure 3. : Comparative statics for STE.

We highlight four comparative statics:

1. Fixed roles ($\gamma = 1$): increasing punishment severity (higher b) lowers $\hat{\delta}$, making it easier to sustain STE.
2. Mild punishment ($b = 0.5$): increasing role uncertainty (lower γ) also lowers $\hat{\delta}$, facilitating STE.
3. Symmetric roles ($\gamma = 0.5$): variation in b has no effect on $\hat{\delta}$ as the cost of deviation

is constant across punishment levels.

4. Asymmetric roles ($\gamma = 0.7$): the effect of b is non-monotonic. When $b < 0.75$, increasing b lowers $\hat{\delta}$. But when $b \geq 0.75$, increasing b raises $\hat{\delta}$.

Overall, the comparative statics reveal that introducing either stronger punishment or role uncertainty individually facilitates strategic communication relative to the baseline. However, when the two factors interact, their joint effect depends on the degree of role uncertainty. Under symmetric roles, stronger punishment has no additional effect on sustainability. In contrast, under asymmetric roles, the effect of punishment becomes non-monotonic, initially promoting cooperation, but ultimately hindering it when the punishment becomes too severe.

To understand this non-monotonicity, it is crucial to recognize the dual role of the parameter b . In the punishment phase (i.e., permanent reversion to BE), the stage-game payoff is $1 - b$ for Sender and b for Receiver. Therefore, b acts as a double-edged sword: increasing b worsens the punishment for Sender but simultaneously makes the punishment state more attractive for Receiver.

Under asymmetric role uncertainty, players face different expected payoffs in the punishment phase. Consider Player A, who acts as Sender with probability $\gamma > 0.5$. Her expected stage-game payoff in the punishment phase is $\gamma(1 - b) + (1 - \gamma)b$. Because she is predominantly Sender, increasing b strictly decreases her expected punishment payoff, thereby increasing her cost of deviation. Conversely, Player B acts as Receiver with probability $\gamma > 0.5$. His expected punishment payoff is $(1 - \gamma)(1 - b) + \gamma b$. For Player B, increasing b actually *increases* his expected payoff in the punishment phase, which lowers his cost of deviation.

A numerical example illustrates how this dynamic shifts the binding constraint. Suppose $\gamma = 0.7$. The continuation value of STE for Player A is $0.7(1 + 0)0.5 + 0.3(1 + 1)0.5 = 0.65$, while for Player B it is 0.85. This indicates that deviating by lying in state R as Sender is intrinsically more tempting for Player A. When punishment is mild ($b = 0.6$), Player A's expected punishment payoff is $0.7(0.4) + 0.3(0.6) = 0.46$, while Player B's is $0.3(0.4) + 0.7(0.6) = 0.54$. Increasing the punishment to a moderate level of $b = 0.7$ lowers Player A's expected punishment payoff to 0.42 while *increasing* Player B's to 0.58. In these two cases, Player A's constraint as Sender in state R binds because she is more tempted to deviate. Increasing b reduces her punishment payoff and helps regulate her deviation. However, when b increases above 0.75, the expected punishment payoff continues to decrease for Player A but increase for Player B (thereby reducing his deterrence), such that Player B's constraint as Sender in state R becomes the binding one, even though his intrinsic temptation is smaller. Therefore, further increasing b only reduces the deviation cost for Player B, making the STE harder to sustain.

C. The Information Rent Equilibrium

In addition to STE, IE can also be sustained in repeated play.⁸ In IE, Sender always tells the truth, and Receiver conditions her response on the state revealed by the mes-

⁸A full analysis is in Appendix A.A1.

sage. When the message indicates aligned interests, Receiver fully follows the message. When it indicates misaligned interests, she responds with the stage-game Nash action, effectively sharing part of the surplus with Sender. Any deviation triggers a permanent reversion to BE.

The key feature of IE is the use of information rent: when the state is R , Receiver plays action M in response to message “ R .” This action, as termed by Wilson and Vespa (2020), gives Sender a positive payoff even in the misaligned state, thereby increasing Sender’s immediate incentive to report truthfully. However, this requires Receiver to forgo some surplus, making IE more demanding in terms of coordination. We assume symmetric strategies across players.

On the equilibrium path, Sender may be tempted to lie in state R to obtain a higher short-run payoff if Receiver were to blindly follow. Similarly, Receiver may deviate by refusing to pay the information rent and instead choosing her myopically optimal action. As in STE, deviations trigger reversion to punishment (action M), yielding off-path payoffs of $(1 - b, b)$. Importantly, in IE, the parameter b not only determines the severity of punishment but also affects the size of the information rent, complicating its overall impact.

IE-BASED COMPARATIVE STATICS. — We conduct a parallel analysis for IE and compute the minimum discount factor $\hat{\delta}$ required to support it under different combinations of b and γ . Figure 4 displays the results. As before, the horizontal axis represents the punishment level b , and the vertical axis represents $\hat{\delta}$. The point in the bottom-left corner with fixed roles ($\gamma = 1$) and mild punishment ($b = 0.5$) serves as our baseline.

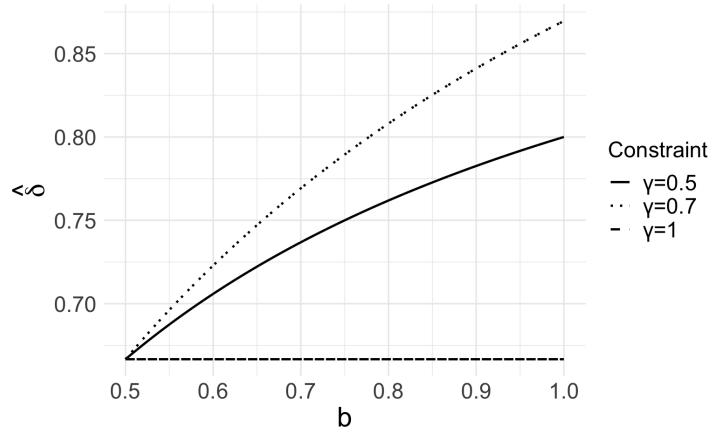


Figure 4. : Comparative statics for IE.

We highlight four comparative statics:

1. Fixed roles ($\gamma = 1$): increasing b has no effect on $\hat{\delta}$. Effects on punishment severity and the size of information rent are canceled out.

2. Mild punishment ($b = 0.5$): increasing role uncertainty (lower γ) has no effect on $\hat{\delta}$. Since Sender and Receiver face symmetric incentives, role uncertainty does not alter the equilibrium condition.

3. Random roles ($\gamma < 1$): increasing b raises $\hat{\delta}$. When roles are not fixed, the binding constraint shifts to Sender who is more likely to be in Receiver role in the future. In this case, a larger b increases the temptation to deviate when acting as Sender in the misaligned state.

4. Non-mild punishment ($b > 0.5$): increasing role uncertainty (lower γ) also raises $\hat{\delta}$. As the chance of being Sender declines, the expected future benefit from truth-telling decreases, while the short-run gain from misreporting remains, making deviation more attractive.

Overall, the $\hat{\delta}$ to support IE is always weakly smaller than that to support STE for any given b and γ . In addition, the comparative statics for IE contrast with those for STE. Introducing either stronger punishment or greater role uncertainty individually has no effect relative to the baseline. However, when the two factors interact, i.e., when punishment is harsh and roles are randomized, their joint effect is detrimental. The interaction increases the incentive to deviate in misaligned states, making IE theoretically harder to sustain with a larger required $\hat{\delta}$.

D. Discussion about the Theoretical Focus

While the repeated game admits a multiplicity of equilibria, we formulate our hypotheses based on the sustainability of STE. We do so not because it is the only truthful and efficient equilibrium, but because STE is strategically simpler. As mentioned in Section II.C, IE theoretically requires weakly smaller $\hat{\delta}$ than STE because the provision of information rent further motivates Sender's truthful report. However, this theoretical robustness comes at the cost of complex strategy. IE requires Receiver to proactively calculate and forgo short-term gains to provide this "information rent," creating a two-sided temptation to deviate. In contrast, STE only requires Sender to resist temptation, while Receiver simply plays her myopic best response. We posit that this strategically simpler STE will serve as a more natural focal point, facilitating easier coordination and thus higher realized efficiency.

Furthermore, our theoretical analysis focusing on truth-telling equilibria implicitly assumes that Receiver's cooperation is automatic once Sender is disciplined. Our hypotheses and empirical analysis will focus on the aggregate efficiency predictions based on the STE benchmark, and then deconstruct the underlying Sender and Receiver behaviors to examine whether these implicit assumptions hold in practice or not.

E. Design and Predictions

We have six experimental treatments (Table 1), varying along two dimensions: punishment severity (b) and role uncertainty (γ). The **Baseline** features fixed roles and mild punishment. Three treatments vary only one factor: **H** (harsh punishment with fixed roles), **UR** (unequal random roles with mild punishment), and **ER** (equal random roles

with mild punishment). The remaining two treatments vary both: **HUR** (harsh punishment and unequal random roles) and **HER** (harsh punishment and equal random roles).

Table 1—: Six treatments.

		Role Uncertainty (γ)		
		Fixed (1)	Unequal Rand. (0.7)	Equal Rand. (0.5)
Punishment (b)	Mild (0.5)	<i>Baseline</i>	UR	ER
	Harsh (0.9)	H	HUR	HER

We measure communication efficiency as $\mathbb{P}(a = L \mid \theta = L)$, defined as the probability Receiver’s action matches the state when interests are aligned. We focus on this specific conditional probability because it serves as a notion of surplus gain in our environment. Specifically, in our stage game, when $\theta = R$, the total surplus is always 1 regardless of Receiver’s action (only the distribution of payoffs varies). When $\theta = L$, the total surplus is 2 if $a = L$, 1 if $a = M$, and 0 if $a = R$. The maximal possible aggregate surplus is gained only when action L is taken in state L . In any truth-telling equilibrium, maximal efficiency of 1 is achieved.

The implemented discount factor was fixed at $\delta = 0.8$ across all treatments. If a treatment’s $\hat{\delta} \leq 0.8$ for a certain equilibrium, then this equilibrium is theoretically sustainable.

Table 2 summarizes the theoretical predictions based on the minimum required discount factors derived in the comparative statics. According to the STE-based analysis, treatments H, ER, and HER can support STE. In contrast, the Baseline, UR, and HUR cannot. The IE-based analysis suggests that all treatments except HUR can support IE.

Table 2—: Predictions under the implemented time discount factor of 0.8.

	<i>Baseline</i>	UR	ER	H	HUR	HER
STE	×	×	✓	✓	×	✓
IE	✓	✓	✓	✓	×	✓

Note: A check mark (✓) indicates that the corresponding equilibrium can be supported. A cross (×) indicates that the equilibrium cannot be supported.

Like any truth-telling equilibrium, both STE and IE are fully efficient: in both equilibria, Sender reveals the state truthfully, and Receiver takes the optimal action for Sender (action L) when the state is L , achieving the maximum possible information transmission efficiency ($\mathbb{P}(a = L \mid \theta = L) = 1$) and lying on the Pareto frontier of the stage game payoffs.

However, precise point predictions are difficult given the multiplicity of equilibria

in repeated games. Following the experimental literature (e.g., Duffy and Heinemann, 2021), we focus on *qualitative treatment effects*. As discussed in Section II.D, the mere existence of multiple efficient equilibria does not guarantee successful coordination. Because STE is strategically simpler and avoids the two-sided temptation inherent in IE, we posit that treatments theoretically supporting STE will facilitate coordination on the efficient outcome more effectively. Consequently, we formulate our hypotheses based on the sustainability of STE as our focal benchmark. The following hypotheses guide our empirical analysis:

Hypothesis 1 (H). *Efficiency in Treatment H is higher than in the Baseline.*

This hypothesis tests the impact of punishment severity under fixed roles. While the Baseline supports only IE, Treatment H supports STE. We predict that the availability of the simpler STE in H will lead to higher realized efficiency.

Hypothesis 2 (UR). *There is no difference in efficiency between the Baseline and UR.*

This hypothesis tests the impact of asymmetric role uncertainty under mild punishment. Since both treatments support IE but not STE, the theoretical benchmark predicts no significant difference in their efficiency ceilings.

Hypothesis 3 (ER). *Efficiency in Treatment ER is higher than in the Baseline.*

This hypothesis tests the impact of symmetric role uncertainty. Unlike the Baseline, Treatment ER theoretically supports STE, predicting a positive treatment effect.

Hypothesis 4 (HUR). *Efficiency in Treatment HUR is lower than in the Baseline.*

This hypothesis tests the interaction of harsh punishment and asymmetric roles. While the Baseline supports IE, Treatment HUR supports neither STE nor IE (only BE is sustainable), predicting a negative treatment effect.⁹

Hypothesis 5 (HER). *Efficiency in Treatment HER is higher than in the Baseline, and not significantly different from Treatment ER.*

This hypothesis tests the interaction of harsh punishment and symmetric roles. While the Baseline supports IE, Treatment HER supports both STE and IE, predicting a positive treatment effect. Since both Treatments HER and ER support STE (and IE), the theoretical benchmark predicts similar efficiency levels.

Importantly, however, efficiency is *not* an identifying statistic for truthful communication or for play of STE. High efficiency can arise even when messages are not truthful if Receivers “trust” in a non-discriminating way, while high Sender honesty need not translate into high efficiency if Receivers do not follow the message in state L . For this reason, we complement efficiency with Sender and Receiver behaviors when interpreting empirical differences between treatments.

F. Alternative Hypothesis based on $\max\{\delta - \hat{\delta}, 0\}$

While our primary hypotheses rely on a binary theoretical criterion that whether the strategically simpler STE is sustainable ($1\{\delta \geq \hat{\delta}\}$), the experimental literature on infinitely repeated games suggests that cooperation rates often depend continuously on the robustness of the equilibrium. Specifically, merely crossing the theoretical threshold does

⁹If we solely focus on STE, then the prediction is no treatment effect.

not guarantee coordination; rather, the difference between the continuation probability and the minimum required threshold ($\delta - \hat{\delta}$) is used as an index to measure the difficulty to sustain certain cooperative equilibrium, and is shown to be positively correlated with realized cooperation (Blonski, Ockenfels and Spagnolo, 2011; Bó and Fréchette, 2018).

To account for this and as suggested by one of the referees, we introduce an alternative continuous metric $\max\{\delta - \hat{\delta}, 0\}$. A value of 0 indicates the equilibrium is unsustainable or is a fragile knife-edge case, while a larger positive value implies greater robustness.

Applying this metric to both STE and IE under our implemented $\delta = 0.8$, we obtain the following values across treatments:

- **Treatment H:** STE ≈ 0.09 ; IE ≈ 0.13 . (Robust for both)
- **Baseline, UR, and ER:** STE = 0; IE ≈ 0.13 . (Robust only for IE; STE is either unsustainable or a knife-edge case)
- **Treatment HER:** STE = 0; IE ≈ 0.02 . (STE is a knife-edge case; IE is only weakly robust).
- **Treatment HUR:** STE = 0; IE = 0. (Neither equilibrium is sustainable)

This continuous measure of equilibrium robustness based on STE and IE generates a distinct, strict ranking of expected efficiency across treatments. It predicts that Treatment H should yield the highest efficiency due to substantial slack in both equilibria. The Baseline, UR, and ER treatments should exhibit intermediate efficiency, driven by the robust IE. HER should perform worse due to its fragile IE slack, and HUR should perform the worst. This leads to our alternative hypothesis:

Alternative Hypothesis. *Efficiency follows the ranking: $H > \{\text{Baseline, UR, ER}\} > \text{HER} > \text{HUR}$.*

III. Experimental Implementation

We conducted the experiment in English using oTree (Chen, Schonger and Wickens, 2016) at the Hong Kong University of Science and Technology (HKUST). Data collection took place in November 2023, and March, April, and November 2025.¹⁰ The design included six treatments, each implemented in seven sessions. Each session involved 12–20 participants, lasting approximately one hour with 7 supergames played.¹¹

A. Subjects and Online Participation

A total of 658 subjects participated in the study, who were undergraduate students recruited from the HKUST subject pool via email invitations with no prior experience in

¹⁰IRB protocol no.: HERP-2023-0307. Project title: Random role assignment in repeated cheap talk. Approval period: 22-Sep-2023 to 22-Sep-2027. Institution: Hong Kong University of Science and Technology.

¹¹Analysis comparing the last 4 supergames to all supergames (see Appendix A.A3) reveals no qualitative differences, indicating that subjects' learning stabilized within this time frame.

this experiment.¹²¹³ We ensured no subject participated in more than one session.

The experiment was conducted online via Zoom with real-time monitoring to ensure strict compliance. Our protocol differs from online experiments using anonymous crowdsourcing platforms in that our subject pool consists exclusively of university students who are proficient in English, cognitively capable, and highly adapted to synchronous online academic environments. Recent methodological work indicates that online experiments with such populations and proper monitoring yield results comparable to traditional laboratory settings (e.g., Arechar, Gächter and Molleman, 2018; Snowberg and Yariv, 2021; Gupta, Rigotti and Wilson, 2023). Consistent with this, we observed high compliance with protocols and no attrition during the sessions.

B. Procedures

To ensure data quality, we adhered to a strict protocol. The session formally began with the experimenter reading aloud an “Opening Message”¹⁴ which explicitly prohibited communication and the use of external devices. Subjects were required to keep their cameras on throughout the session. The experimenter continuously monitored the video feed to ensure subjects were attentive and situated in a private environment. In the rare event of technical issues (e.g., camera malfunction), the experimenter provided immediate assistance via private chat.

Subsequently, to ensure valid pairing, we verified that an even number of participants (minimum 12) were present. In cases where an odd number of subjects attended, we solicited a volunteer (or randomly selected a participant) to withdraw in exchange for the full show-up fee.

Once the session was finalized, the experimenter shared the screen and read the instructions aloud to establish common knowledge of the game rules.¹⁵ Subjects were permitted to ask questions during this phase. Following the instructions, subjects received a link to the oTree interface. They were required to review the instructions themselves and pass a comprehension quiz with 100% accuracy prior to a non-incentivized practice game. The practice game mirrored the official interface to ensure familiarity with the game mechanics. Seven official games followed after the practice game.

C. Implementation of the Repeated Game

ROLE ASSIGNMENT. — In treatments with *fixed roles*, subjects were randomly assigned as either Senders or Receivers at the start of the session. At the beginning of each su-

¹²We recruited a total of 772 participants. However, seven sessions for Treatment HUR contained a programming error regarding the role uncertainty parameter (implemented as 50-50 rather than 70-30). We corrected this error and re-ran seven sessions. The analysis in this paper relies exclusively on data from the correct sessions. Qualitative patterns in the excluded data were similar to the reported results.

¹³**Caveat: We identified a procedural issue in the data collection. We plan to re-run in-lab sessions under the correct protocol to replace the data, but believe the qualitative patterns are robust. Comparisons between the existing data and the re-run data will be conducted.**

¹⁴The Opening Message is provided in Appendix A.A2.

¹⁵Sample instructions are provided in Appendix A.A2.

pergame, Senders were randomly re-matched with Receivers. Both the pairing and the roles remained fixed for the duration of the supergame. In treatments with *random roles*, subjects were assigned generic player IDs (A or B). Pairs were formed randomly at the start of each supergame and remained fixed. However, roles were randomly assigned at the start of *each round*: Player A acted as Sender and Player B as Receiver with probability γ ; Player A acted as Receiver and Player B as Sender with probability $1 - \gamma$.

DISCOUNT FACTOR AND SESSION STRUCTURE. — We visualized the continuation probability of $\delta = 0.8$ using a spinning wheel interface (80% green, 20% red) at the end of each round to induce a vivid feeling of the “shadow of the future.” If the wheel stopped on green, the game continued; otherwise, it terminated. Both players in a pair observed the same spin outcome.

To standardize the total session duration and synchronize participant progress, we pre-generated the random termination draws at the beginning of each session. Specifically, in each session, the seven supergames had lengths of $\{2, 3, 4, 5, 6, 7, 8\}$ rounds. The order of these lengths was randomized for each treatment but fixed across all pairs and sessions within that treatment.¹⁶

D. Payoffs

Each payoff unit in Figure 1 corresponded to 40 HKD in the experimental interface. The average earning was approximately 160 HKD (≈ 20 USD) for a session of 60-80 minutes, which included a 40 HKD show-up fee. Payments were made electronically via the university’s Autopay System.

IV. Aggregate Empirical Findings

To ensure reliable statistical inference, we employ two approaches. Our *primary analysis* relies on pooled mean comparisons with standard errors clustered at both the unique player A and unique player B indices.¹⁷ As a *robustness check*, we perform mean comparisons using session-level averages via Mann-Whitney rank-sum tests.¹⁸ In the following subsections, reported significance levels are based on the primary analysis. Unless otherwise specified, the robustness checks yield consistent significance results.

We present our findings by synthesizing the aggregate efficiency results with the underlying Sender and Receiver behavior. This approach allows us to directly evaluate the hypotheses formulated in Section II.

¹⁶This is the aforementioned procedure mistake. This procedure does not correspond to the continuation rule described to subjects, and thus violates the no-deception principle, though non-intentional. We are re-running in-lab sessions under the correct protocol. We will replace the data upon completion. Results in the current draft with the existing data are provisional.

¹⁷Our main text only displays figures using data in all rounds of all supergames (except for efficiency). Three additional sets of figures using data in all rounds of last four supergames, round-1 of all supergames, round-1 of last four supergames, are displayed in Appendix A.A3. All figures show similar patterns as figures in the main text, indicating no evidence of first-round difference or learning effects.

¹⁸Four sets of figures using data in all rounds of all supergames, all rounds of last four supergames, round-1 of all supergames, round-1 of last four supergames, are displayed in Appendix A.A4.

A. Fixed Roles with Harsh Punishment

We first examine the impact of punishment severity in the absence of role uncertainty (Hypothesis 1). Figure 5 displays the aggregate efficiency across treatments,¹⁹ while Figures 6 and 7 illustrate the corresponding behavior. In these figures, the horizontal axis displays six treatments, paired by the degree of role uncertainty (e.g., the Baseline is paired with Treatment H; both treatments have fixed roles but differ in punishment severity). The significance level (based on two-way clustered standard errors) for pair-wise comparisons are denoted by significance stars (*) over the corresponding brackets (***) denotes $p < 0.01$, ** denotes $p < 0.05$, and * denotes $p < 0.1$).

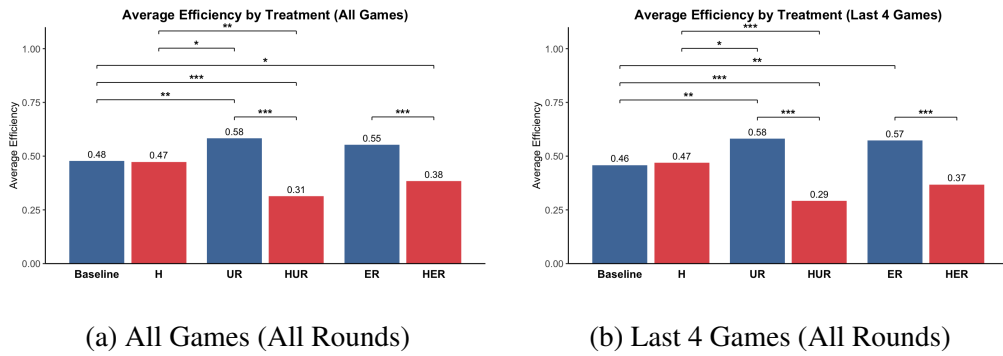


Figure 5. : Average Efficiency by Treatment.

Result 1. *Introducing stronger punishment **does not** promote efficiency compared to the Baseline.*

This finding contradicts Hypothesis 1, which predicts that the availability of STE in Treatment H would lead to higher efficiency.²⁰ Decomposing this result reveals a striking asymmetry in how harsh punishment affects players.

¹⁹It is informative to compare our results in Baseline with the “Partners” treatment in Wilson and Vespa (2020). While the game structures are similar, the parameters differ: we use a discount factor of $\delta = 0.8$ and a payoff parameter of $b = 0.5$, whereas Wilson and Vespa (2020) use $\delta = 0.75$ and payoffs corresponding to $b \simeq 0.67$ (where Receiver gets 2/3 and Sender 1/3 under the Partial action). Empirically, we observe higher efficiency levels in our Baseline (for all games, 55% in the first round as shown in Appendix and 48% in all rounds) compared to their Partners treatment (for all games, 43% in the first round and 32% in all rounds). The higher efficiency in our setting was likely facilitated by the higher continuation probability.

²⁰Our findings in Treatment H are closely related to the robustness treatment “Partners-R2” in Wilson and Vespa (2020). In their setting, reducing Sender’s babbling payoff to zero successfully increased efficiency to 58.6%. However, our Treatment H ($b = 0.9$) achieved an overall efficiency of 47%. As discussed in Section I, comparing these empirical outcomes provides highly complementary insights. Wilson and Vespa (2020) employ an elegant asymmetric punishment that strictly punishes Sender while keeping Receiver’s payoff from the safe action constant. In contrast, our parameter b introduces a structural punishment: increasing b to 0.9 harshly punishes Sender but simultaneously makes the safe action highly lucrative for Receiver. This higher opportunity cost of trusting triggers the “safety trap” we observe in this subsection, where Receivers retreat to the safe option and nullify Sender’s honesty.

On Sender side, harsh punishment is highly effective. As shown in Figure 6 (Left Panel), Senders in Treatment H exhibit a significantly higher truth-telling frequency in state R than in the Baseline by 21% ($p < 0.001$), which is in line with the theoretical incentive constraints.²¹ In the aligned state L (Right Panel), truth-telling remains near the ceiling in both treatments, confirming that the treatment effect is isolated to the conflict state and that Senders generally understand the incentives.

However, this increased honesty is effectively nullified by Receiver behavior. As illustrated in Figure 7 (Panel a), Receivers in Treatment H do not reciprocate with higher trust in message “ L ”. Instead, they exhibit a conservative form of distrust. While distrust in the Baseline is characterized by “noisy” rejection (high action R , see Panel e), distrust in Treatment H is concentrated on the safe option (action M , see Panel c), which is significantly higher than in the Baseline by 18% ($p < 0.001$). This conservatism is pervasive: even when receiving message “ R ”, Receivers in Treatment H default to action M (Panel d, $p < 0.001$). Furthermore, since the frequencies of choosing action M in response to messages “ L ” and “ R ” are similar in Treatment H, it appears unlikely that Receivers are applying an “information rent” strategy. Consequently, the potential efficiency gains from increased honesty are nullified by conservative Receivers trapped in a safe action M .

It is important to note that this “safety trap” is likely driven by more than simple risk aversion. While risk aversion naturally increases the attractiveness of the safe option, the sheer magnitude of the shift points to a fundamental change in incentives. The increased payoff for Receiver by action M ($b = 0.9$) drastically raises the *opportunity cost* of trust. For Receiver, the safe action offers a high, immediate, and certain payoff, creating a tension between immediate maximization and long-run efficiency. This structural tension overwhelms Sender’s discipline, as the safe action becomes attractive for Receivers if her belief about Sender’s truthfulness is not high enough.

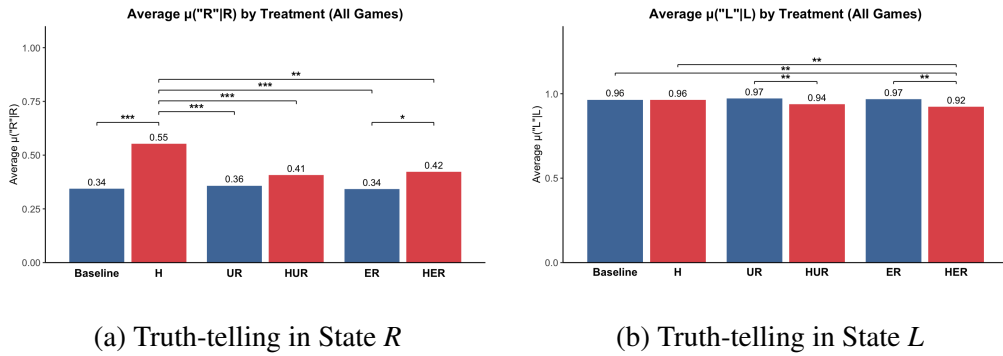


Figure 6. : Sender Behavior by Treatment (All Games).

²¹Although Sender truthfulness in state R in Treatment H is significantly higher than in the Baseline overall, the across-round trend figures in the Appendix A.A6 reveal that this gap gradually narrows over time.

B. Role Uncertainty with Mild Punishment

Next, we evaluate the impact of role uncertainty under mild punishment (Hypotheses 2 and 3).

Result 2. *Introducing asymmetric role uncertainty **promotes** efficiency compared to the Baseline, though this result has limited robustness.*

Result 3. *Introducing symmetric role uncertainty **promotes** efficiency compared to the Baseline, specifically in the last four games.*

We find that introducing role uncertainty tends to improve efficiency relative to the Baseline. In the asymmetric role uncertainty (UR) treatment, the primary analysis indicates a positive effect of 10% ($p = 0.035$), contradicting the null prediction of Hypothesis 2, although this result is insignificant in the session-level robustness check (see Appendix). For symmetric role uncertainty (ER), the effect is insignificant in all games but becomes significant in the last four games ($p = 0.041$), providing partial support for Hypothesis 3.

Decomposing this efficiency gain reveals an asymmetric effect of role uncertainty on Sender and Receiver. As shown in Figure 6 (Left Panel), role uncertainty does not significantly alter Sender truth-telling in state R compared to the Baseline. Instead, role uncertainty induces an increase in Receiver trust in message “ L ”. Figure 7 (Panel a) shows that Receivers in UR (and in the last four games of ER) are significantly more likely to choose action L in response to message “ L ” ($p = 0.013$ and $p = 0.031$, respectively). Furthermore, Panels c and d indicate that while Receivers in UR and ER exhibit a similar frequency of action M in response to “ L ” as the Baseline, they choose action M in response to “ R ” significantly more often than the Baseline ($p = 0.018$ and $p = 0.001$, respectively). This implies that Receivers are more willing to pay an “information rent” in UR and ER.

This pattern suggests a phenomenon that under mild punishment, Receivers are willing to trust imperfect signals or pay information rents to sustain cooperation, a behavior that emerges specifically when roles are randomized with mild punishment.

C. Role Uncertainty with Harsh Punishment

Finally, we analyze the interaction of harsh punishment and role uncertainty (Hypotheses 4 and 5).

Result 4. *Introducing asymmetric role uncertainty and harsh punishment **reduces** efficiency compared to the Baseline.*

Result 5. *Introducing symmetric role uncertainty and harsh punishment **reduces** efficiency compared to the Baseline, though this result has limited robustness; furthermore, adding harsh punishment **reduces** efficiency compared to ER.*

Consistent with Hypothesis 4, Treatment HUR yields significantly lower efficiency than the Baseline by 17% ($p = 0.002$). However, contrary to Hypothesis 5, which predicted that HER would improve efficiency similar to ER, Treatment HER performs sig-

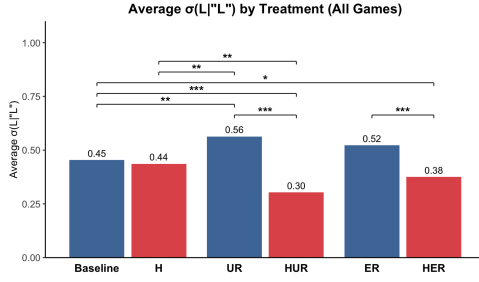
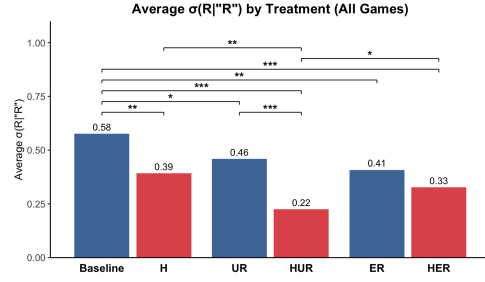
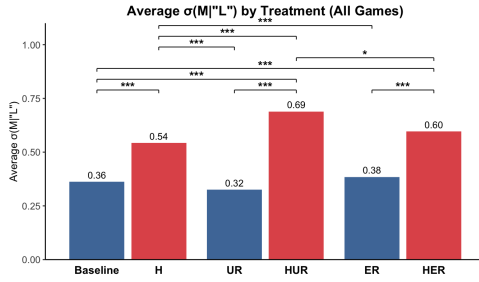
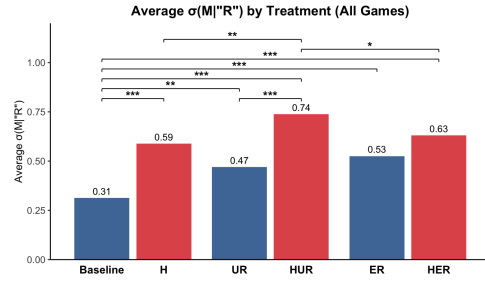
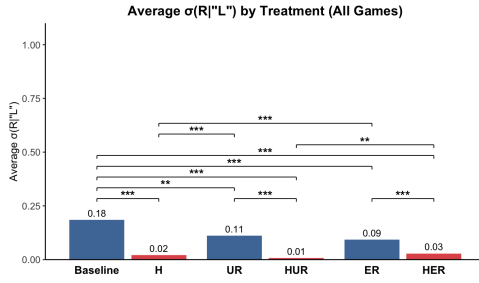
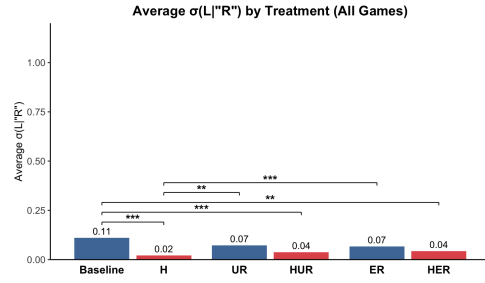
(a) Action L | Msg “ L ” (Trust)(b) Action R | Msg “ R ” (Follow)(c) Action M | Msg “ L ” (Safe)(d) Action M | Msg “ R ” (Rent/Safe)(e) Action R | Msg “ L ” (Distrust)(f) Action L | Msg “ R ” (Confusion)

Figure 7. : Receiver Behavior by Treatment (All Games). Left column: Responses to Message “ L ”. Right column: Responses to Message “ R ”.

nificantly worse than the Baseline by 10% ($p = 0.087$, though insignificant in last four games and also insignificant with the session-level test) and significantly worse than ER by 17% ($p = 0.002$).

The mechanism behind this efficiency drop differs fundamentally from the theoretical prediction. Theory suggests that under HUR, efficiency fails because Senders' incentives to lie are not effectively regulated. However, Figure 6 shows that Senders in HUR and HER are *more* truthful than in the Baseline in state R , though this is only significant for HER in the last four games ($p = 0.038$). In addition, Senders in HER are more truthful than those in ER ($p = 0.086$, but insignificant in the session-level test), yet the efficiency in HER is lower than ER.

Instead, the failure is driven by a collapse in Receiver trust. Comparing the Baseline to HUR and HER in Figure 7 (Panel a), we see that the interplay of role uncertainty and harsh punishment significantly reduces trust in response to message “ L ” by 15% and 7% ($p < 0.001$ and $p = 0.1$, respectively). Relatedly, the choice of action M (Panel c) increases significantly in HUR and HER by 33% and 24% (both with $p < 0.001$) compared to the Baseline. Compared to ER, HER also exhibits significantly lower trust in message “ L ” by 14% ($p = 0.004$) and a higher frequency of action M in response to “ L ” by 22% ($p < 0.001$). Moreover, in HUR and HER, the frequency of action M in response to “ R ” is similar to that in response to “ L ”, suggesting it is unlikely that Receivers are applying an “information rent” strategy.

This reveals a negative interaction effect: while role uncertainty under mild punishment fosters trust, role uncertainty under harsh punishment amplifies the “safety trap.” Even though Senders are more honest, this environment drives Receivers to abandon cooperation in favor of action M . This “trust collapse” occurs regardless of whether role uncertainty is symmetric (HER) or asymmetric (HUR).

Continuing the discussion of the “trust collapse”, beyond simple risk aversion, role uncertainty may further erode the relational restraint that sustains trust. When roles can reverse, within-round payoff asymmetries may feel less morally salient because they are partially offset *ex ante* by future role realizations. In our harsh-punishment environment, where the intermediate action yields a very high sure payoff ($b = 0.9$), this weakening of immediate inequality concerns can make the safe option normatively and strategically easier to justify, contributing to the greater observed collapse in trust.

V. Supergame-Level Analysis: Micro-foundations of Treatment Effects

The aggregate findings in Section IV reveal significant deviations from the STE-based hypotheses. First, harsh punishment successfully disciplines Sender but fails to elicit Receiver trust. Second, mild punishment combined with role uncertainty triggers unconditional trust despite Sender dishonesty. Third, the interaction of harsh punishment and role uncertainty leads to a collapse in trust despite higher Sender honesty.

These aggregate puzzles challenge two core implicit assumptions embedded in our theoretical benchmark. First, standard theory implicitly assumes that once Sender's incentive problem is resolved (e.g., via harsh punishment), Receiver will automatically cooperate and trust. Our findings in the harsh punishment treatments directly contradict

this. Second, the STE-based hypotheses assume that high efficiency must stem from strict truth-telling. Yet, the high efficiency observed under mild punishment with role uncertainty suggests an alternative pathway to cooperation. This occurs even though Treatment ER, which theoretically supports STE, deviates from truth-telling. In short, the aggregate data reveals that efficiency can collapse despite high truthfulness, or thrive despite low truthfulness.

To understand the mechanisms driving these discrepancies, we must look beyond treatment averages. While the aggregate data highlights asymmetric effects on Senders and Receivers, it remains unclear how these outcomes emerge from the joint distribution of strategies within a supergame. We need to determine whether the treatments merely alter the frequency of certain action profiles (the intensive margin) or if they induce entirely different action profiles (the extensive margin).

To explore this, we proceed to a subject-supergame level analysis by visualizing the behavioral profiles of pairs in each supergame using scatterplots. In all panels of Figures 8 to 10, the horizontal axis (x -axis) represents the (average) Sender's truth rate in state R , while the vertical axis (y -axis) represents the (average) Receiver's trust rate in response to message "L". The size of each dot is proportional to the frequency of supergames exhibiting that specific behavioral profile, and the percentages located at the corners and along the edges indicate the share of observations falling exactly on those extreme points or boundaries. For fixed-role treatments (Figure 8), each dot plots the exact truth and trust rates of a fixed pair. For random-role treatments (Figures 9 and 10), each dot represents the *pair centroid*: the average truth and trust rates of the two players in a dyad, with crosshair arms extending from the centroids to visualize the within-pair heterogeneity across the two dimensions.

Mapping these coordinates to theoretical strategies provides an intuitive geometric interpretation of the outcomes. The four corners of the unit square correspond to distinct behavioral archetypes. The bottom-left corner, $(0,0)$, corresponds to BE, where Senders are completely deceptive and Receivers are entirely distrusting. Conversely, the top-right corner, $(1,1)$, represents the truth-telling equilibria, characterized by full honesty and full trust. The off-diagonal corners capture asymmetric behaviors: at the top-left corner, $(0,1)$, Receivers fully trust despite Senders being completely deceptive; whereas at the bottom-right corner, $(1,0)$, Senders are fully truthful but Receivers refuse to trust. Furthermore, the diagonal space is highly informative. Observations lying above the 45-degree line indicate that a pair's trust rate exceeds their truth rate, visually capturing the phenomenon of trust despite imperfect messages. Conversely, observations below the 45-degree line imply that truthfulness exceeds trust, reflecting a state of reduced trust despite more informative messages.

A. Mechanism 1: The Double-edged Sword of Harsh Punishment (Fixed Roles)

We first examine why the theoretical prediction failed in Treatment H despite the observed increase in Sender truthfulness. Figure 8 compares the behavior profile distribution of the Baseline and Treatment H.

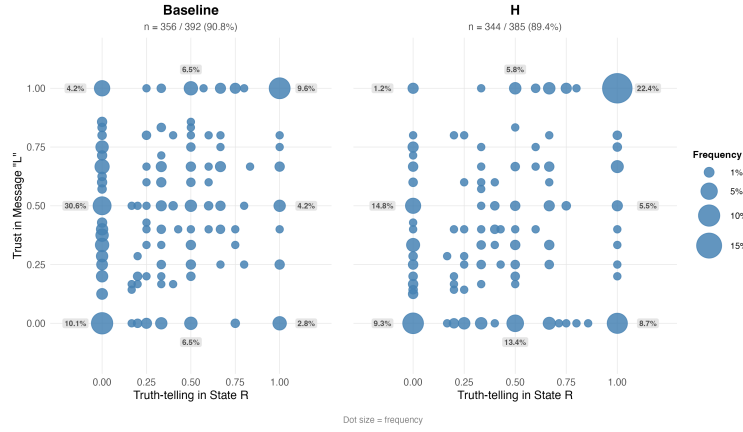


Figure 8. : Fixed-role Treatments: Baseline vs. H

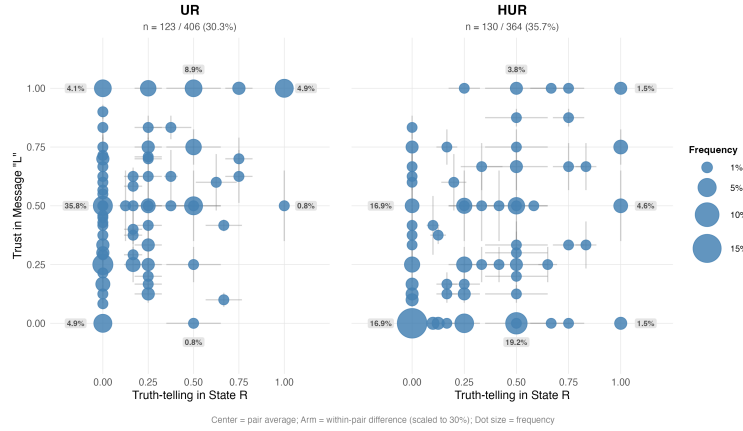


Figure 9. : Asymmetric Random-role Treatments: UR vs. HUR



Figure 10. : Symmetric Random-role Treatments: ER vs. HER

Notes: The figures display supergame-level behavior profiles across treatments. In **fixed-role treatments** (Baseline, H), each dot represents a supergame, plotting Sender's truth rate in state *R* (x-axis) against Receiver's trust rate to message "L" (y-axis). In **random-role treatments** (ER, UR, HER, HUR), each dot represents the average truth and trust rates for each paired players within a supergame. For these treatments, crosshair arms indicate the within-pair heterogeneity in each dimension, scaled by a factor of 0.3 to reduce visual clutter. Dot size is proportional to frequency. Percentages by the corners indicate the share of observations located exactly at the corners; percentages along the edges indicate the share of observations on the corresponding boundary (excluding corners). The sample includes only supergames where all constituent rates are observed; sample sizes and retention rates are reported at the top of each panel.

Finding 1. *In the Baseline, behavior is characterized by low Sender honesty and scattered Receiver trust. In Treatment H, harsh punishment successfully disciplines Senders, yet it triggers a bifurcation in Receiver response. These countervailing shifts explain the absence of aggregate efficiency gains.*

In the Baseline (Figure 8, Left Panel), the data concentrates on the left half of the unit square and exhibits a vertical dispersion. A significant mass (44.9%) of Senders adopt a strategy of complete deception, clustering at Sender truth rate of 0. Given any level of Sender truthfulness, Receivers' trust rates are vertical scattered, consistent with a noisy babbling equilibrium where trust is speculative rather than responsive to information.

In contrast, Treatment H (Figure 8, Right Panel) reveals a structural polarization. The entire distribution migrates rightward, confirming that harsh punishment ($b = 0.9$) effectively disciplines Senders. However, this improved honesty drives two opposing outcomes. On one hand, the fraction of pairs with full trust to full honesty (top-right corner) doubles from 9.6% to 22.4%. On the other hand, the fraction of pairs with zero trust to non-zero honesty (x -axis and the bottom-right corner) increases from 9.3% to 22.1%.

This bifurcation illustrates the double-edged nature of harsh punishment: while it enables high-efficiency coordination for some, the larger opportunity cost of trust drives Receivers to deviate from trust to action M . Consequently, the efficiency gains from the former are offset by the collapse in trust of the latter.

This result exposes the fragility of a core implicit assumption in STE: that Receiver's participation is automatic once Sender's incentive constraint is satisfied. Our data reveals that this assumption does not hold when the outside option is attractive. In Treatment H, even though Senders comply with the discipline imposed by harsh punishment, Receivers are not the passive beneficiaries assumed by theory. Instead, the high value of action M ($b = 0.9$) increases Receiver's opportunity cost of trust, causing them to reject Sender's honesty. Thus, the failure of efficiency in H is not a failure of disciplining Sender, but a failure to satisfy Receiver's enhanced need for safety.

B. Mechanism 2: The Blind Trust Equilibrium with Mild Punishment (Random Roles)

We next investigate the unexpected Receiver-driven high efficiency observed in treatments with mild punishment and random roles (ER and UR). Figures 9 (Left Panel) and 10 (Left Panel) display the behavior profiles for these treatments.

Finding 2. *Under mild punishment with role uncertainty, behavior is characterized by high Receiver trust despite low Sender honesty, a pattern consistent with BTE that leads to improved efficiency. This is visualized as a concentration of observations above the 45-degree line, where Receiver trust exceeds Sender truthfulness.*

The scatterplots for UR and ER reveal a behavioral pattern: a significant cluster of supergames located above the 45-degree line in the unit square, indicating that players' trust as Receivers is greater than their truthfulness as Senders. Compared with Baseline, the increased trust in UR and ER is not built on Sender truth. Notably, a large share of pair centroids hover near $x = 0$ and $y > 0$ (39.9% in UR and 39.1% in ER), representing players being fully deceptive as Sender while trusting to some degree as Receiver.

The relatively short crosshair arms on these points indicate that this is not an artifact of averaging polarized behaviors, but rather a shared strategy where both players lie when acting as Sender but trust when acting as Receiver.

We identify this broad behavioral profile as the empirical manifestation of BTE.²² BTE only emerges in environments with role uncertainty. In BTE, both players lie as Sender but fully trust as Receiver, sharing the deceptive gains as Sender on the equilibrium path.

BTE emerges as a focal point due to two reasons. First, BTE achieves efficiency 1 as action L is always taken and lies at the Pareto-frontier. Specifically, BTE is payoff-equivalent to STE: in STE, players are truthful as Senders and always extract maximal payoff 1 as Receiver, in the misaligned state the “losses” are incurred when the player is Sender while the “gains” are incurred when the player is Receiver. With symmetric role uncertainty, players A and B get the same expected payoff 0.75 in each round. With asymmetric role uncertainty, players A gets expected payoff 0.65 and B gets 0.85 in each round. In BTE, things are reversed. In the misaligned state the “losses” are incurred when the player is Receiver while the “gains” are incurred when the player is Sender. With symmetric role uncertainty, players A and B get the same expected payoff 0.75 in each round. With asymmetric role uncertainty, players A gets expected payoff 0.85 and B gets 0.65 in each round.

Second, BTE has trivial requirement on δ to be supported. As Sender is getting her desired action on the path, she does not have incentives to deviate. Only Receiver who needs to blindly trust has other candidate options (e.g., switching to BE). When punishment is mild ($b = 0.5$), a risk-neutral Receiver is myopically indifferent between blindly trusting message “ L ” (expected payoff 0.5) in BTE and playing action M (which yields payoff $b = 0.5$) in BE.

Crucially, this indifference highlights two distinct advantages of BTE over STE for Treatments UR and ER. First, BTE does not rely on the strict behavioral assumption that a Receiver’s trust must be conditioned on Sender’s truthfulness. Second, BTE requires the lowest possible critical discount factor ($\hat{\delta}$), thereby achieving maximal robustness in terms of $\max\{\delta - \hat{\delta}, 0\}$.

To understand the first advantage, consider that STE inherently demands a conditional form of trust: a Receiver’s willingness to trust depends entirely on the belief that Sender is disciplined and honest. In contrast, under mild punishment, BTE eliminates this implicit assumption. Because the opportunity cost of blind trust is zero, Receivers do not need to rely on the premise that Sender must be honest to avoid a loss. Thus, high efficiency emerges in these treatments not because Senders are disciplined (as STE would require), but because Receivers face no downside risk, rendering the complex check on Sender honesty unnecessary.

The second advantage becomes evident when evaluating equilibrium robustness through the lens of $\max\{\delta - \hat{\delta}, 0\}$. In treatments ER and UR, STE is either a fragile knife-edge case (with $\max\{\delta - \hat{\delta}, 0\} = 0$) or theoretically unsustainable. It is therefore theoretically intuitive that STE fails to drive the efficiency gains in these treatments, as subjects naturally struggle to coordinate on an equilibrium with zero margin for error. In con-

²²For the full analysis, see Appendix A.A5.

trust, BTE requires only a trivial discount factor ($\hat{\delta} = 0$) and achieves the largest possible distance between the implemented discount factor and the minimum required threshold.

Consequently, in the data we see players trusting each other even though messages are not perfectly informative. The empirical evidence aligns with the theoretical intuition: the fragile STE does not make a difference under its own strict requirements, while the highly robust BTE successfully organizes behavior and drives the observed efficiency gains.

C. Mechanism 3: Trust Collapse and Role Detachment under Harsh Punishment

Finally, we analyze the breakdown of cooperation in treatments combining harsh punishment with role uncertainty (HER and HUR). We contrast the Right Panels of Figures 9 and 10 with their mild-punishment counterparts.

Finding 3. *In HUR, the interaction of harsh punishment and role uncertainty precipitates a collapse in trust, reducing efficiency. The clusters at the top-left quadrant disappear and those at the top-right corner shrink, consistent with the theoretical prediction that neither BTE nor STE can be supported. Relatedly, observations migrate from high-trust regions to the bottom region, displaying a “trust collapse” and creating a mode clustered at the origin (BE). Though few, a profile of “honest skepticism” emerges at the bottom-right corner, where players always tell the truth yet refuse to trust.*

Finding 4. *In HER, the interaction of harsh punishment and role uncertainty yields a nuanced breakdown. While the persistence of clustering at the top-left and top-right corners indicates that a fraction of supergames still sustain BTE or truth-telling equilibria, overall trust declines. Similar to HUR, observations in HER gravitate toward the origin (BE) in a “trust collapse,” with a quantitatively small “honest skeptic” profile emerging at the bottom-right.*

Figures 9 and 10 visually confirm a pronounced phenomenon of “sinking trust.” Unlike the clusters above the 45-degree line observed in ER and UR, the distribution in HUR and HER concentrated below the negative diagonal line (where either high trust is met with low truth, or high truth is met with low trust), collapsing vertically toward the bottom axis. While Senders truth rate slightly shifts rightward, this honesty is effectively decoupled from Receiver behavior.

The drop in trusting behavior is mainly driven by the increased opportunity cost of trust compared to the choice of action M . Unlike BTE in Treatments UR and ER, under harsh punishment ($b = 0.9$), BTE is no longer a “costless” strategy for Receiver.

While blind trust still yields an immediate expected payoff of 0.5, the safe action M now yields 0.9. This drastic shift transforms BTE from a strategy supported by myopic indifference to one that is heavily Receiver-constrained. To sustain BTE in HER, Receiver must be willing to forgo the instant, dominant gain from the safe action in exchange for future continuation values, requiring a much stricter discount factor ($\hat{\delta} = 0.62$, for details, see Appendix A.A5). In HUR, this constraint becomes impossible to satisfy for the player more likely to be Receiver. Consequently, without the cushion of indifference, Receiver’s incentive constraint binds tightly, and the “blind trust” mechanism

collapses under the weight of the high opportunity cost.

On the other hand, trusting becomes harder in STE as well. Though Senders are more truthful, myopically in each stage game, Receiver needs to hold a belief that message “L” is at least 90% truthful (requiring Sender truth rate to be 0.89) so as to trust instead of choosing action M . Consequently, for positive degrees of truth rate that are below 0.89, we observe an increase of points lying along the x -axis from 0.8% in UR to 19.2% in HUR and from 1.7% in ER to 18.2% in HER.

In addition, for the points with higher than 0.89 truth rate (basically along the edge $x = 1$), though few, we observe a higher fraction of full trust among them in UR (4.9% out of 5.7%) and ER (4.6% out of 6.9%) compared to HUR (1.5% out of 7.6%) and HER (5.8% out of 10.3%).

Moreover, while the dominant outcome is a complete collapse to babbling, we observe an emerging, albeit small, profile of “honest skepticism” in HUR (1.5%) and HER (1.9%) at the bottom-right corner, where players are honest as Senders yet completely distrustful as Receivers. Although quantitatively small, this profile theoretically illustrates a unique failure of cross-role strategic consistency, or “role detachment.” Under harsh punishment, these subjects appear to optimize strictly within their current role frame rather than internalizing their future potential roles. As Senders, fear of the harsh penalty induces compliance with truth-telling; yet as Receivers, facing the high opportunity cost of trust, fear of betrayal drives them to action M . Consequently, a coordination failure arises where honesty is offered but never reciprocated, highlighting how high stakes can undermine the strategic reasoning required for reciprocal strategies.

D. The Dynamics of Distrust: Timing and Triggers of the Safe Action

The figures in the previous subsections illustrate the aggregate behavioral profiles within supergames. However, to fully understand the nature of the “safety trap” and the breakdown of cooperation, it is necessary to examine the dynamics of Receiver behavior. Specifically, we ask: when Receivers choose the safe action M in response to message “L”, is it a calculated punishment triggered by a Sender’s previous deviation (as standard grim-trigger strategies assume), or is it an unprovoked, myopic response?

To address this, we focus on the subset of supergames where message “L” is sent at least once. Figure 11 plots the distribution of the round in which a Receiver *first* chooses action M in response to “L”. The horizontal axis denotes this exact round of first occurrence, and the gray bar captures the fraction of supergames where action M is *never* chosen.²³

Finding 5. *In treatments with mild punishment or fixed roles (Baseline, UR, ER, H), a substantial fraction of supergames (30%-40%) never witness a breakdown of trust into action M . In stark contrast, in HUR and HER, the mode shifts to Round 1, indicating that Receivers default to distrusting message “L” from the onset of the interaction.*

Figure 11 reveals a fundamental shift in Receiver behavior under the interaction of

²³The gray bars in Figure 11 align with the efficient equilibria in which action M should never be taken when message is “L”.

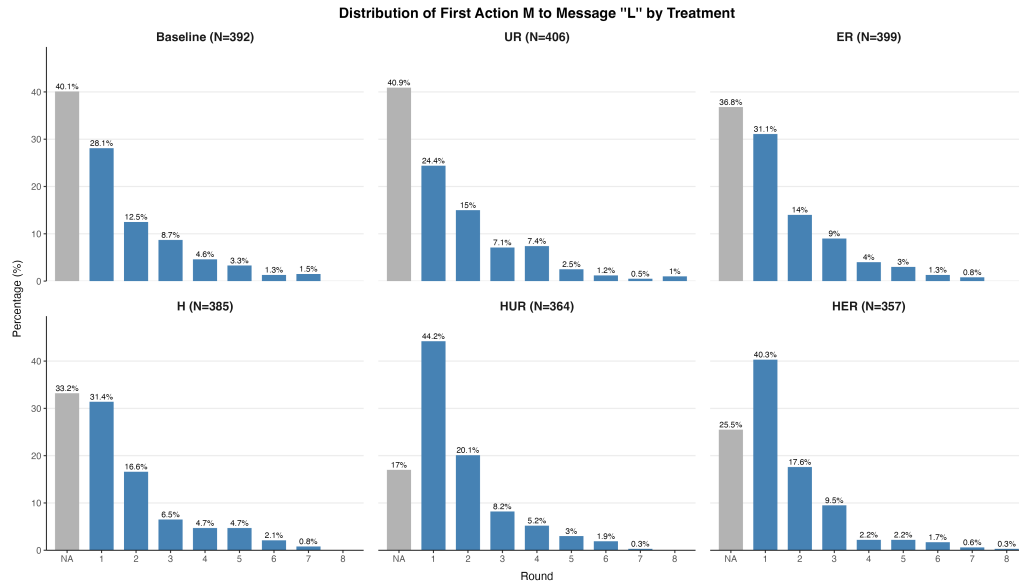


Figure 11. : Distribution of the First Action M to Message "L" by Treatment.

Notes: The figure displays the distribution of the round in which a Receiver first chooses the safe action M in response to message "L", conditional on message "L" being sent at least once in a supergame. The horizontal axis represents the round number of this first occurrence. The gray bar indicates the fraction of supergames where action M is never chosen in response to message "L" throughout the entire interaction.

harsh punishment and role uncertainty. In HUR and HER, over 40% of the first instances of action M occur in the very first round. This implies that the “trust collapse” identified in Finding 4 is not a consequence of failed attempts at cooperation over time. Instead, driven by the high opportunity cost of a mistrust ($b = 0.9$), Receivers exhibit a myopic retreat to the safe option without affording Senders an opportunity to establish a truthful reputation.

This immediate retreat suggests that the safe action serves less as a conditional punishment device and more as an absorbing state driven by strategic uncertainty. The pattern is consistent with the fragility of cooperation in high-stakes environments: when the outside option is highly attractive ($b = 0.9$), the basin of attraction for the cooperative equilibrium shrinks. Consequently, even minor fluctuations in Receiver’s second-order beliefs (beliefs about Sender’s intent) can trigger a shift toward the risk-dominant safe action, appearing as an “unprovoked” abandonment in the observable history.

To further dissect the motivations behind the safe action, we trace the history back exactly one round prior to the *first* instance where a Receiver chooses action M in response to message “ L ”. The horizontal axis in Figure 12 details the exact outcome of this preceding round using a four-part nomenclature, formatted as “[Same/New Sender]-[State][Message][Action]”. For example, a label such as “Y-LLL” indicates that the current Sender was the previous Sender (denoted by “Y”), the realized state was L , the message sent was “ L ”, and Receiver’s chosen action was L . Based on these granular outcomes, the brackets in Figure 12 group the prior-round histories into five overarching categories from the left to the right of the figures: (1) *Initial-round* (“Init”), where the first action M occurs in Round 1 and thus lacks prior history; (2) *Mutual-cooperation*, where the previous round featured truthful communication and trusting responses (e.g., state L , message “ L ”, action L ; or state R , message “ R ”, action R or M); (3) *Sender-defied*, where Sender lied in either state L or R ; (4) *Receiver-defied*, where Receiver distrusted either message “ L ” or “ R ”; and (5) *Both-defied*, where Sender lied and Receiver simultaneously distrusted.

Finding 6. *Across all treatments, the majority of the first transitions to the safe action are non-punitive. All treatments except UR have over 70% of the first action M to message “ L ” are triggered either at the initial round or immediately following a round of mutual cooperation.*

As shown in Figure 12, the *Initial-round* is the dominant category across all treatments. When examining ongoing rounds, either *Sender-defied* (which justifies action M as a punishment for Sender deviation) or *Mutual-cooperation* (which represents an unprovoked breakdown of trust) is the second most frequent trigger.

Notably, in all treatments, the fraction of unprovoked breakdowns following mutual cooperation is substantial (ranging from 17.8% to 26.3%), comparable to or even exceeding the punitive responses. Only Treatment UR have moderately more punitive trigger (*Sender-defied*, 30.4%) than the unprovoked breakdown.

This pattern highlights a fundamental fragility in sustained communication. It demonstrates that even when pairs successfully coordinate on cooperation, Receivers frequently abandon it abruptly. This unprovoked deviation is not a retaliatory response to betrayal,

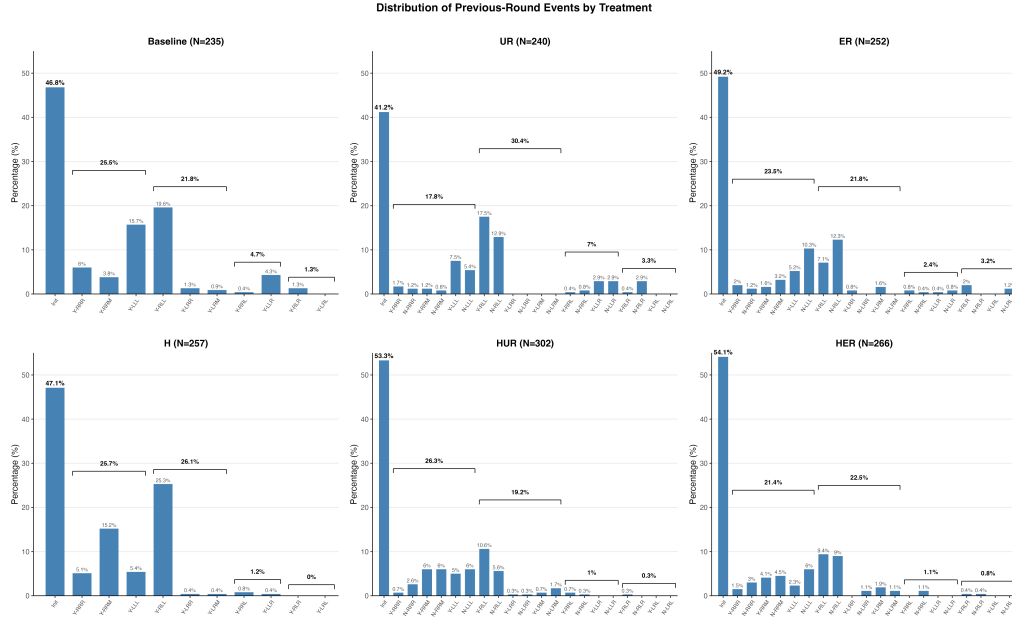


Figure 12. : Distribution of Previous-Round Types Prior to the First Action $M|L$.

Notes: The figure categorizes the history exactly one round prior to the first instance a Receiver chooses action M in response to message “ L ”. The horizontal axis details the exact outcome of the preceding round using the format “[Same/New Sender]-[State][Message][Action]”, where the prefix indicates whether the current Sender was the previous Sender (“Y”) or not (“N”). The brackets group these fine-grained outcomes into five overarching categories: *Initial-round* (occurs in Round 1, thus lacking prior history), *Mutual-cooperation*, *Sender-defied* (Sender lied in the previous round), *Receiver-defied* (Receiver distrusted in the previous round), and *Both-defied*.

but rather highlights the instability of the cooperative equilibrium under harsh punishment. The high value of the safe action implies that cooperation is no longer robust to strategic noise or trembling beliefs. Receivers may abandon cooperation not because they observed a defection, but because the strategic risk of maintaining trust becomes prohibitive relative to the secure outside option.

E. Conclusion of Micro-Level Analysis

Our micro-level decomposition speaks to the aggregate puzzles by analyzing supergame-level behavior profiles and their within-game dynamics. The unchanged aggregate efficiency in Treatment H is characterized by a bifurcation: while some pairs coordinate on truth-telling equilibria, a significant fraction migrate to the “safety trap,” where Sender honesty is nullified by Receiver conservatism. Conversely, the unexpected success of ER and UR is driven by the emergence of BTE. Under mild punishment ($b = 0.5$), Receiver’s myopic indifference between trusting and the action M relaxes the incentive constraint ($\delta \geq 0$), making BTE a robust and efficient focal point where mutual tolerance substitutes for strict requirements on honesty.

In contrast, the interaction of harsh punishment and role uncertainty (HER and HUR) proves detrimental. In HUR, BTE cannot be supported. In HER, while theoretically possible, BTE loses its focalness as Receiver’s incentive to deviate to the safe action M increases, requiring a prohibitive discount factor ($\delta > 0.62$). Instead of cooperation, we observe a massive migration toward the babbling equilibrium, accompanied by a qualitative emergence of “honest skepticism” (role detachment), where harsh punishment enforces Sender compliance but undermines Receiver trust. Crucially, our dynamic analysis reveals that this undermined trust is not primarily a calculated punishment for past deception. Receivers either refuse to trust from the initial round or abruptly abandon established cooperation without provocation, trapping pairs in an inefficient, risk-averse absorbing state of unreciprocated honesty.

VI. Conclusion

This paper evaluates the institutional determinants of information transmission in repeated interactions. While standard theory suggests that expanding the “shadow of the future” via harsh punishment or role uncertainty should facilitate fully revealing equilibria, our experiment documents a fundamental asymmetry. We find that these mechanisms operate distinctly on Senders and Receivers, leading to equilibrium selection outcomes that defy refinements based solely on Sender incentives.

Our primary contribution is identifying the “safety trap” under harsh punishment. While severe sanctions successfully strengthen Sender’s truth-telling incentives, they inadvertently hinder Receiver’s incentives to trust. By increasing the opportunity cost of trust, harsh punishment drives Receivers toward the safe option, crowding out the truth-telling equilibrium. This result provides a behavioral caveat to the folk theorem in communication games: increasing punishment severity is not monotonic in efficiency because it creates a wedge between honesty and trust.

In contrast, efficiency is improved under mild punishment and role uncertainty through the BTE. Unlike truth-telling equilibria, BTE relies on a heuristic of mutual forbearance. The prospect of role reversal, combined with the low opportunity cost of trust, fosters a permissive environment where Receivers accept imperfect signals. This dynamic suggests that in professional networks, such as “revolving door” relationships, cooperation is often sustained not by strict monitoring, but by a form of reciprocal tolerance that persists despite dishonesty.

However, the interaction of harsh punishment and role uncertainty precipitates a breakdown in strategic consistency, manifesting as “role detachment.” Subjects decouple their behavior across roles, adhering to the truth as Senders to avoid punishment while refusing to trust as Receivers. This suggests that high stakes may dismantle the cognitive empathy required for reciprocal strategies, as players optimize strictly within their current role frame rather than internalizing the supergame strategy.

Our dynamic analysis of within-game play further elucidates these failures by challenging the traditional interpretation of the middle action as a conditional punishment device. We find that the retreat to the safe option is rarely a retaliatory response to a Sender’s deception. Instead, particularly under harsh punishment, distrust manifests either as an immediate, myopic reaction from the very first round, or as an unprovoked abandonment of established cooperation. This reveals that the safe action functions less as a grim-trigger enforcement mechanism and more as an absorbing state driven by strategic uncertainty, fundamentally altering how we should model relationship breakdowns in repeated communication.

These findings offer a novel perspective on the trade-off between honesty and efficiency. Fully revealing equilibria, such as STE, rely on “high-tension” enforcement: they require Receiver to trust *only because* they have verified that Sender faces severe consequences for lying. This demands high cognitive sophistication and patience. Conversely, the BTE observed under random roles with mild punishment operates on a “low-tension” principle. By keeping the punishment severity low, the environment renders Receiver myopically indifferent between trusting and playing safe. This indifference facilitates cooperation by removing the burden of verification from Receiver. Consequently, when roles are random, we face a stark design dichotomy: high sanctions aim for honesty but risk collapsing into the “safety trap,” while mild sanctions preserve cooperation but only by tolerating the dishonesty inherent in BTE.

This dichotomy presents a critical challenge for institutional design when the objective is strictly truthful reporting rather than mere social surplus. Since simply adjusting the severity of punishment cannot simultaneously solve the “safety trap” and eliminate BTE, future research must explore mechanisms that decouple these incentives. An important open question is how to design institutions that elicit honesty in environments with role uncertainty without dismantling the trust required for cooperation. Potential avenues may involve separating Sender’s penalty from Receiver’s outside option, or introducing external auditing third-parties. Such designs would enforce discipline without burdening Receiver with the risk of policing, thereby allowing honesty to coexist with the robustness of low-tension trust.

Ultimately, these insights imply that the optimal design of communication institutions depends heavily on the specific nature of the desired outcome. In settings where deal flow and surplus are paramount, such as investment syndication, moderate penalties that foster a BTE-like tolerance may be sufficient. However, in environments where integrity is non-negotiable, such as peer review or judicial oversight, relying on harsh sanctions can be counterproductive. In these cases, our results suggest that stability cannot be achieved simply by ramping up the stakes for the participants; instead, it requires structural interventions that insulate the fragility of trust from the necessary harshness of discipline.

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APPENDIX

A1. Derivation of the Minimal Required Time Discount Factor $\hat{\delta}$ for Information Rent Equilibrium

We consider a generalized version of the Information Rent Equilibrium (IE), which may not always be feasible but is theoretically well-defined. In this setting, the on-path strategy is: *truth-telling as Sender*, and *message-following with information rent r when the truthful message is “R” as Receiver*. We analyze four possible one-shot deviations.

CASE 1. — Player A (as the current Sender) deviates from truth-telling to lying in state R .

- No deviation:

$$r + \frac{\delta}{1-\delta} \left(\gamma \frac{1+r}{2} + (1-\gamma) \frac{2-r}{2} \right)$$

- Deviation:

$$1 + \frac{\delta}{1-\delta} (\gamma(1-b) + (1-\gamma)b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2-2r}{4-3r-2b-3\gamma+2\gamma r+4\gamma b}$$

CASE 2. — Player B (as the current Sender) deviates from truth-telling to lying in state R .

- No deviation:

$$r + \frac{\delta}{1-\delta} \left((1-\gamma) \frac{1+r}{2} + \gamma \frac{2-r}{2} \right)$$

- Deviation:

$$1 + \frac{\delta}{1-\delta} ((1-\gamma)(1-b) + \gamma b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2-2r}{1-r+3\gamma+2b-4\gamma b-2\gamma r}$$

CASE 3. — Player A (as the current Receiver) deviates by refusing to pay the rent on a truthful message “R”.

- No deviation:

$$1 - r + \frac{\delta}{1 - \delta} \left(\gamma \frac{1+r}{2} + (1 - \gamma) \frac{2-r}{2} \right)$$

- Deviation:

$$1 + \frac{\delta}{1 - \delta} (\gamma(1 - b) + (1 - \gamma)b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2r}{2 + r - 3\gamma - 2b + 2\gamma r + 4\gamma b}$$

CASE 4. — Player B (as the current Receiver) deviates by refusing to pay the rent on a truthful message “R”.

- No deviation:

$$1 - r + \frac{\delta}{1 - \delta} \left((1 - \gamma) \frac{1+r}{2} + \gamma \frac{2-r}{2} \right)$$

- Deviation:

$$1 + \frac{\delta}{1 - \delta} ((1 - \gamma)(1 - b) + \gamma b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2r}{-1 + 3r + 3\gamma + 2b - 2\gamma r - 4\gamma b}$$

These four inequalities jointly determine the minimal discount factor $\hat{\delta}$ needed to sustain the generalized IE. The two equilibrium types discussed in the main text—IE and STE—are special cases of this framework:

- **IE:** $r = 1 - b$ (Receiver pays the information rent)
- **STE:** $r = 0$ (Receiver always plays the myopic best response)

SPECIAL CASE: IE ($r = 1 - b$) The inequalities simplify as follows:

- Case 1:

$$\delta \geq \frac{2b}{1 + b - \gamma + 2\gamma b}$$

- Case 2:

$$\delta \geq \frac{2b}{3b + \gamma - 2\gamma b}$$

- Case 3:

$$\delta \geq \frac{2 - 2b}{3 - 3b - \gamma + 2\gamma b}$$

- Case 4:

$$\delta \geq \frac{2 - 2b}{2 + \gamma - b - 2\gamma b}$$

SPECIAL CASE: STE ($r = 0$) In this case, Receiver has no incentive to deviate, as they always choose the myopic best response. Only Sender's constraints matter:

- Case 1:

$$\delta \geq \frac{2}{4 - 2b - 3\gamma + 4\gamma b}$$

- Case 2:

$$\delta \geq \frac{2}{1 + 2b + 3\gamma - 4\gamma b}$$

COMPUTING $\hat{\delta}$ For any given (γ, b) pair, the maximum of these inequalities yields the minimal required discount factor $\hat{\delta}$. The resulting values for STE and IE are illustrated in Figure 3 and Figure 4, respectively.

A2. Sample Instructions for Treatment HUR

Opening Message:

- You should stay in a quiet place now.
- Please turn on your webcam for the entire experiment.
- Please choose the “speaker view” option on your top-right corner.
- You are currently muted. You need my permission to unmute yourself.
- You can send me a (private) chat message if you have any questions.
- Please change your name to two letters + one number, e.g. WY5.
- Make sure that you are using your laptop or desktop now.
- Please do NOT use the Internet Explorer. Most of other web browsers (Chrome, Safari, Firefox, Edge, and Opera) are fine.
- Mobile phone or tablet PC (like iPad) may create some technical issue.

Formal Instructions:

Welcome to the experiment! This experiment studies decision-making in pairs of individuals. Please read the instruction carefully. Please do not communicate with other participants during the experiment. The experiment will last about 60 minutes. The payment you will receive from this experiment depends on your decisions.

1. Overview of this experiment

You will go through the following modules:

- **Instruction:** you will learn about the basic environment of this experiment.
- **Quiz:** it will give you an opportunity to learn whether your understanding about the instruction is correct or not.
- **One Practice Game:** it will allow you to be familiar with the interface of the game.
- **Seven Official Games:** the formal part of the experiment.

2. Your session and pair

There are XX participants in this session, including yourself. At the beginning of the experiment, one-half of the participants will be assigned with a player ID “A” and the other half with a player ID “B”. Your player ID will remain fixed throughout the experiment.

The experiment consists of 7 official games. At the beginning of each game, one player A and one player B are randomly paired. The pair is fixed within each game. After each game, participants will be reshuffled to form new pairs. Everyone won’t know the identity of the paired participant.

3. Overview of each game: roles, states and choices

Each game consists of at least 1 round, possibly multiple rounds of interactions.

At the beginning of each round, there are two possibilities of the role assignment: “A is Sender, B is Receiver” (with 70% chance) or “B is Sender, A is Receiver” (with 30% chance). This means that the player A is more likely to be Sender in each round.

At the beginning of each round, there are two possibilities of the state: Left or Right. One of the states will be randomly determined by the computer. They are equally likely.

If your role in the current round is Sender, you will observe the state. Then you will be asked to choose a message to send to Receiver: “State is left” or “State is right”. You are allowed to send any of these two messages regardless of the actual state being Left or Right.

If your role in the current round is Receiver, you will not observe the state. Instead, you will receive a message from Sender in your pair. Then you will be asked to choose an action: Left, Middle, or Right.

Main takeaway: you will possibly play multiple rounds in each game. You will interact with the same participant within the game. But the state (Left or Right) and your roles

(Sender or Receiver) will be random in each round. The participant with player ID “A” will have more chance to be Sender in each round.

4. Overview of each game: round-reward

The reward you can get in each round depends on the current state and the current Receiver’s action, which can be summarized in the table below. Each cell corresponds to the reward for a given state and action. Remember that in each cell, no matter whether you are player A or B, the first element denotes the current Sender’s round-reward, while the second element denotes the current Receiver’s round-reward. One unit denotes 1HKD.

For example, in the cell located by “State is Right, Receiver’s Action is Left”, Sender’s round-reward is 40, and Receiver’s round-reward is 0.

If you are the current Sender in a round, your round-reward will be 40 if Receiver’s action is Left, no matter the state. It will be 4 if Receiver’s action is Middle, no matter the state. It will be 0 if Receiver’s action is Right, no matter the state.

If you are the current Receiver in a round, your round-reward will be 40 if your action correctly matches the state. It will be 36 if your action is Middle, no matter the state. It will be 0 if your action is the opposite to the state.

Table A1—: Payoff distribution.

		Receiver’s action		
		Left	Middle	Right
State	Left	40,40	4,36	0,0
	Right	40,0	4,36	0,40

When the game terminates, your total reward of each game is the sum of all round-rewards.

5. Overview of each game: information feedback

At the end of each round, you will be provided with a summary of what happened in this round, including the state, Sender’s chosen message, Receiver’s chosen action, and your reward in this round.

Within each game, during each round, no matter you are Sender or Receiver, you can see the history information about the previous rounds (if exist), including the past role assignments, past states, past messages and the past actions.

The following is a constructed example of the available history at round 4: we have the information about the past three rounds.

For instance, we can know from the list that in round 2, the realized state was Left, the role assignment was “A was Receiver, B was Sender”, the message from Sender (played by B) was “State is Left”, and the action from Receiver (played by A) was Left.

6. Random termination of each game

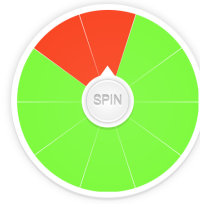
Table A2—: Example of history information.

Round	State	Sender	Message	Action
1	Right	A	“State is Right”	Right
2	Left	B	“State is Left”	Left
3	Right	A	“State is Right”	Right

As mentioned, each game has at least one round, possibly multiple rounds. After each round, there are two possibilities: the game terminates, or it continues to the next round.

The probability of going to the next round is 80%, while the probability of termination is 20%. You will see a spinning wheel that consists of red area (20%) and green area (80%) as illustrated below. Once you click the “Spin” button, the wheel starts spinning. If the spinning wheel stops at the green area, the game continues. Otherwise, the game terminates. Both participants in the same pair will see the same outcome from the wheel for each round.

Note that no matter how many rounds the game has been played, the probability of continuation is always 80%.



7. Summarizing the uncertainties in each game

In the previous pages, you have learnt the details of the game, including your possible roles, available choices and the corresponding rewards. Basically, you will play 7 official games. In each game, you may have multiple rounds. In each round, the current Sender sends a message after observing the State, and the current Receiver takes an action after receiving the message.

In each round of the game, there are three aspects of uncertainty: random role assignment, random state and random termination.

At the beginning of each round, roles (Sender or Receiver) are randomly assigned with an unequal chance (if you are player A, you have 70% chance to be Sender; if you are player B, you have 30% chance to be Sender), and a state (Left or Right) is randomly determined with a 50-50 chance. At the end of each round, the game will continue to the next round with 80% chance.

8. Your cash payment

The practice game won't affect your cash payment. After completing the last official game, the computer will randomly select one game out of 7 games for your payment. Every game has an equal chance to be selected for your payment, so it is in your best interest to take each game equally seriously.

For example: Suppose the randomly selected game ended at the 5th round and the round-rewards for you were 36, 40, 40, 40, 0, then your final cash payment is 40HKD show-up fee plus the total reward of the selected game 156HKD, which in total is 196HKD.

9. Quiz and practice game

Next, you will first finish a short quiz. You can only proceed with all correct answers. Then, you will participate in a practice game. The official games will follow afterward.

If you have any confusion about the process of the experiment, you can review the instruction. If you are ready for the quiz, please press "NEXT PAGE".

Quiz

Please answer the following 8 questions:

- 1) During the whole experiment, you will be paired with the same participant in all games. (True or False)
- 2) In each game, you may be Sender for several rounds and be Receiver for other rounds, depending on the computer's randomization. If you are player B, you are more likely to be Sender than your paired participant. (True or False)
- 3) In each game, it is possible to have only one round. (True or False)
- 4) Sender, but not Receiver, will learn the current state determined by the computer. (True or False)
- 5) Within a game, before making decision, you will know the past decisions in previous rounds, but not the past states. (True or False)
- 6) Your final cash payment may depend on your paired participant's decision. (True or False)
- 7) Look at the round-reward table. If the state is Left and Receiver's action is Right, what will be Sender's round-reward? (40, 36, 4, 0)
- 8) Your future roles (Sender or Receiver) may affect your total reward within each game. (True or False)

The 1 Practice game

You have passed the quiz! Before participating in the 7 official games, you will have 1 practice game which won't affect your cash payment. It's for letting you be familiar with the experiment's interface and the process of a game. Once the practice game is over, it moves to the official games.

The 7 official games

After the last game, the experiment will be over. You will be instructed to fill in the receipt for your payment. The amount you earn will be paid electronically via the HKUST Autopay System to the bank account you provide to the Student Information System (SIS). The Finance Office of HKUST will arrange the auto-payment. An email notification will be sent to your HKUST email address on the pay date under the name of Sender “FOPSAP” (Finance Office Payment System Auto Payment).

A3. Supplementary Figures with Two-way Clustering

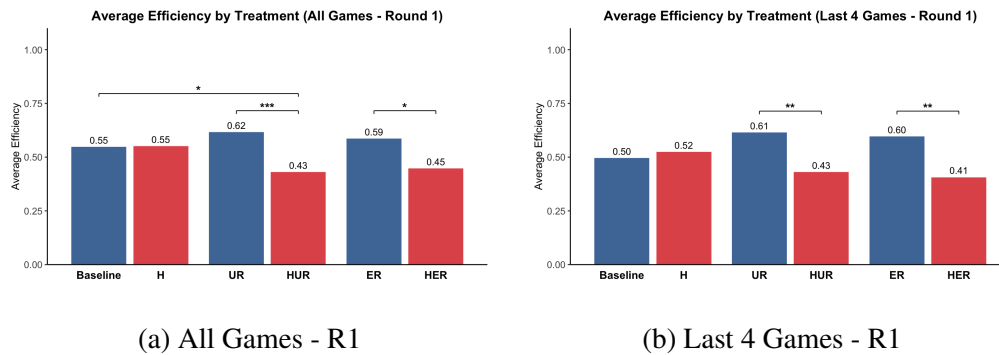
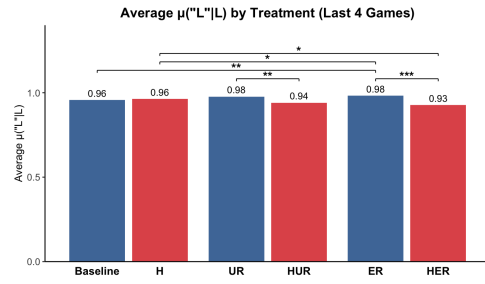
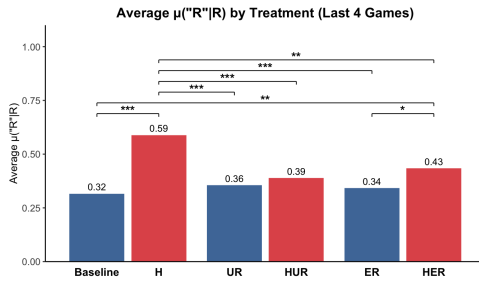
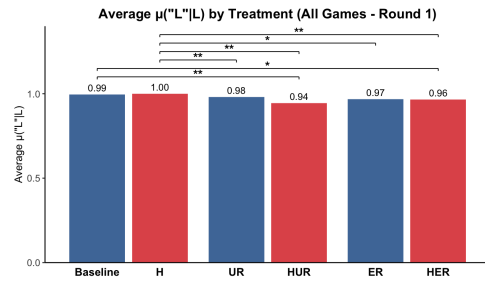
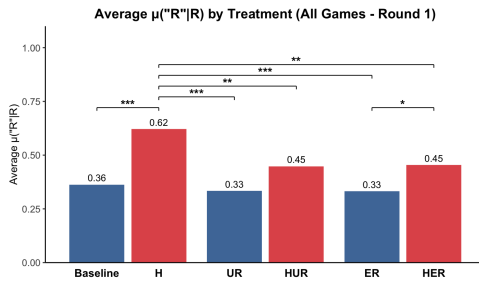


Figure A1. : Supplementary Efficiency Comparison (Two-way Clustering).

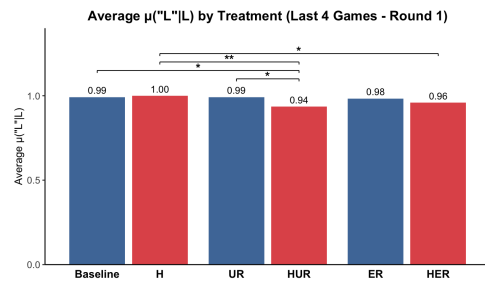
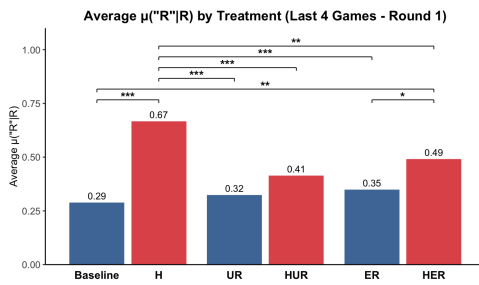


(a) Truth in R (Last 4 Games - All Rounds) (b) Truth in L (Last 4 Games - All Rounds)



(c) Truth in R (All Games - R1)

(d) Truth in L (All Games - R1)



(e) Truth in R (Last 4 Games - R1)

(f) Truth in L (Last 4 Games - R1)

Figure A2. : Supplementary Sender Behavior (Two-way Clustering).

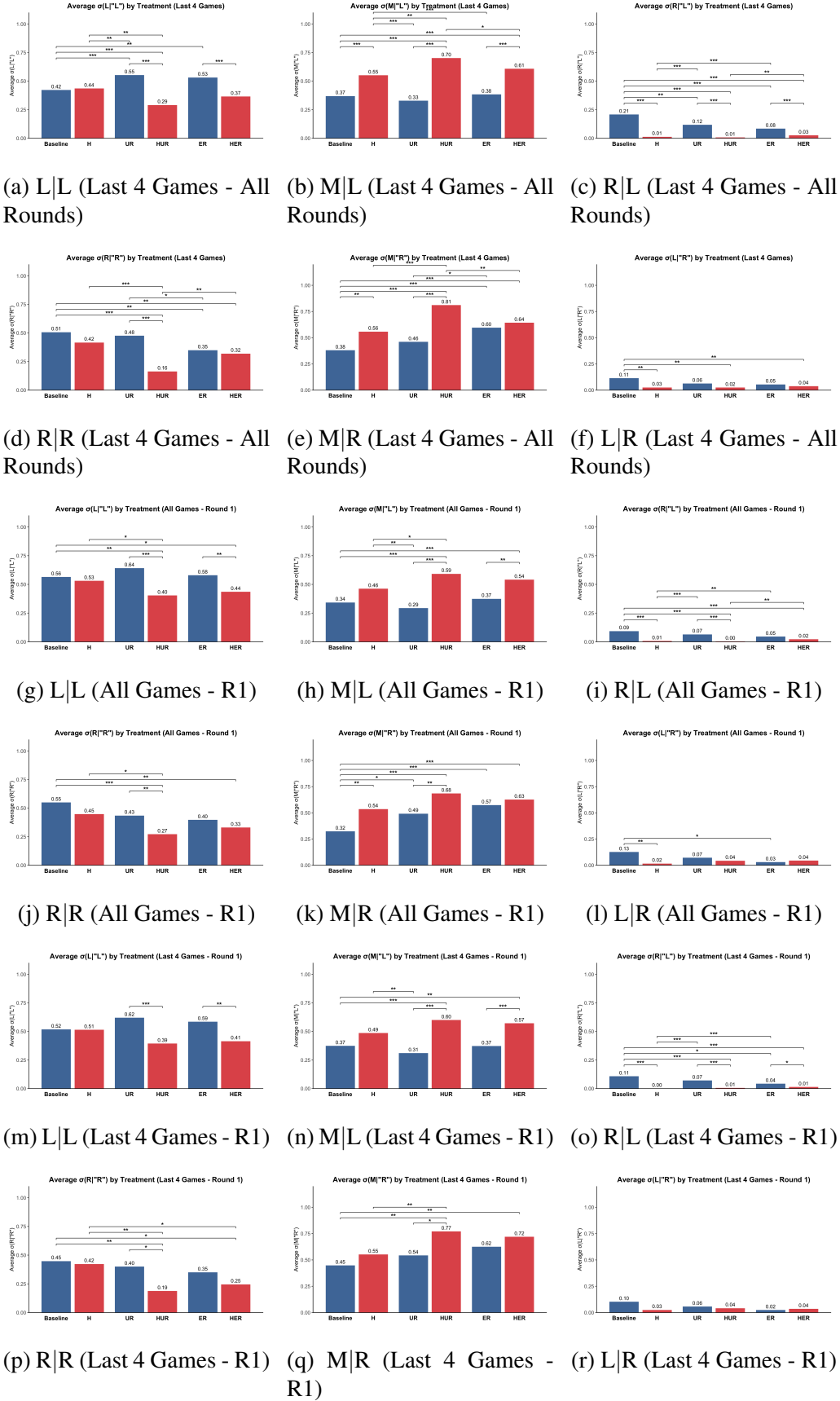
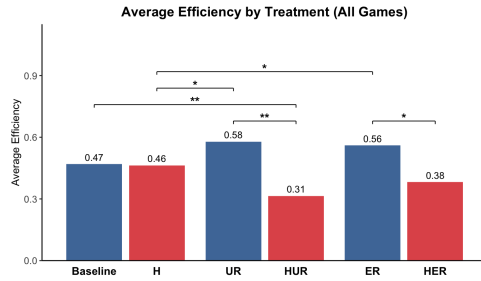
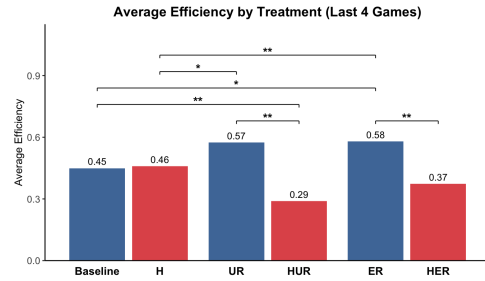


Figure A3. : Supplementary Receiver Behavior (Two-way Clustering).

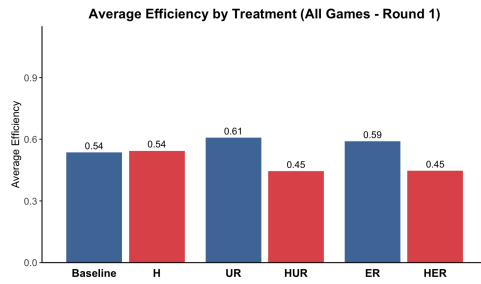
A4. Supplementary Figures with Session-level Ranksum Tests



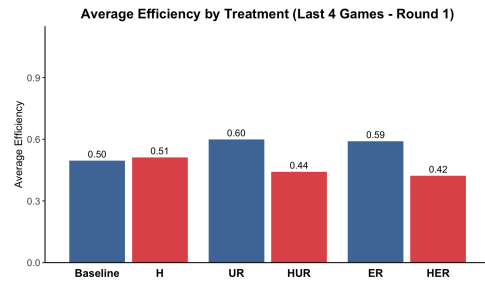
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

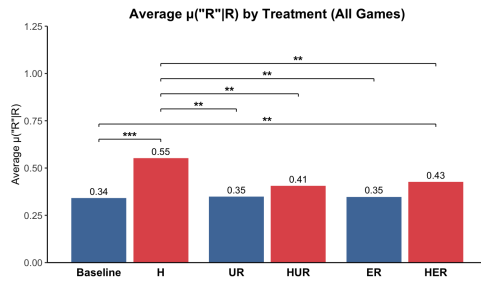


(c) All Games - R1

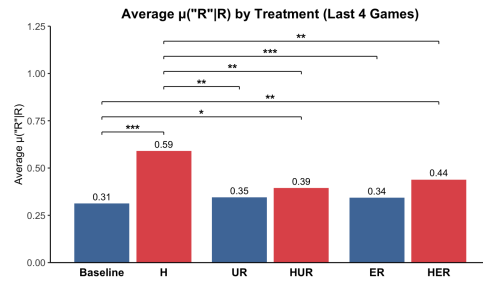


(d) Last 4 Games - R1

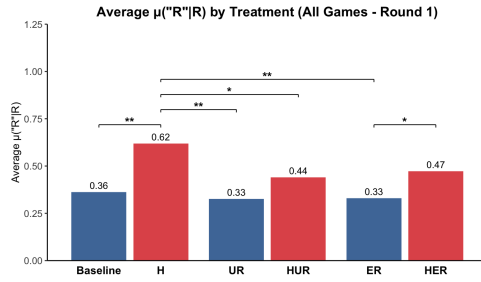
Figure A4. : Efficiency (Session-level Ranksum).



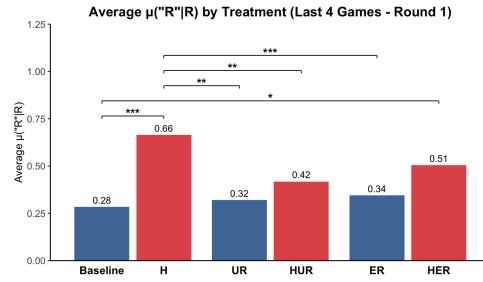
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

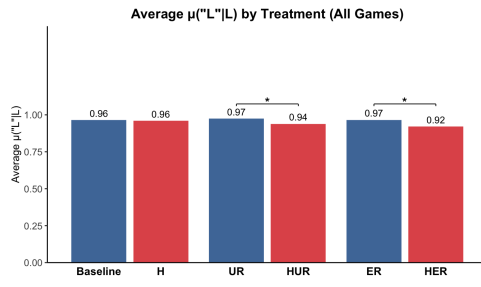


(c) All Games - R1

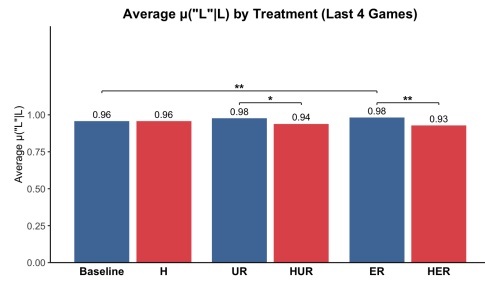


(d) Last 4 Games - R1

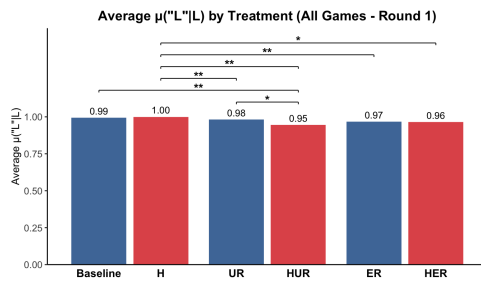
Figure A5. : Sender Behavior: Truth-telling in State R (Session-level Ranksum).



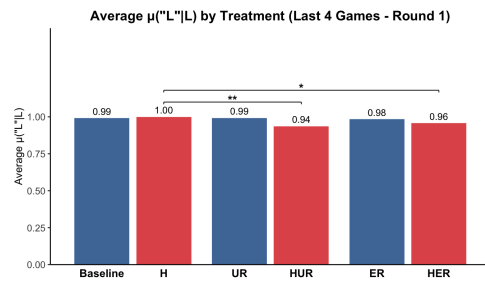
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

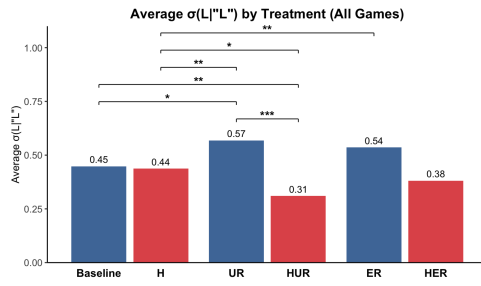


(c) All Games - R1

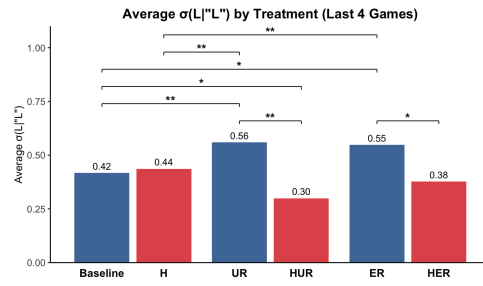


(d) Last 4 Games - R1

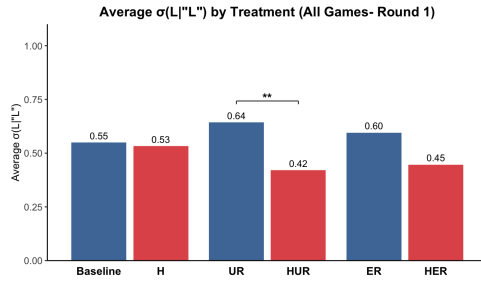
Figure A6. : Sender Behavior: Truth-telling in State L (Session-level Ranksum).



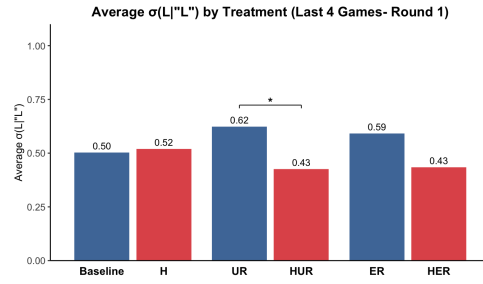
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

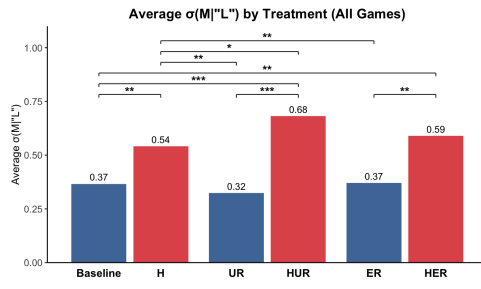


(c) All Games - R1

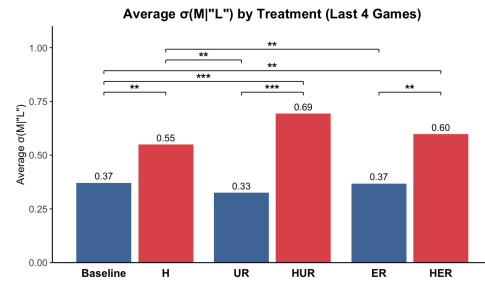


(d) Last 4 Games - R1

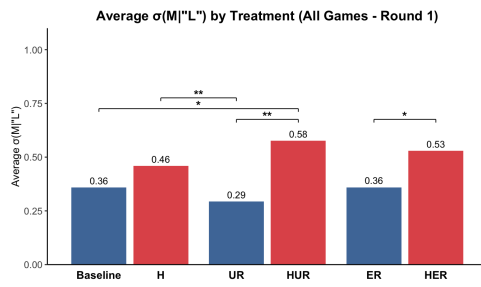
Figure A7. : Receiver Behavior: Action L given Message " L " (Session-level Ranksum).



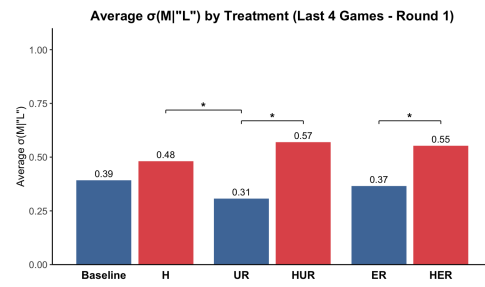
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

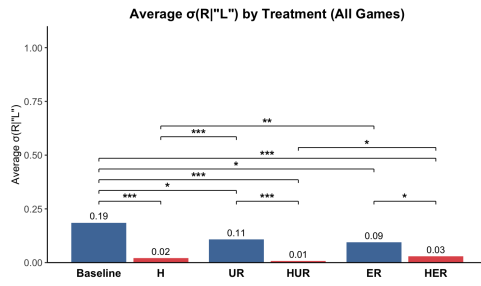


(c) All Games - R1

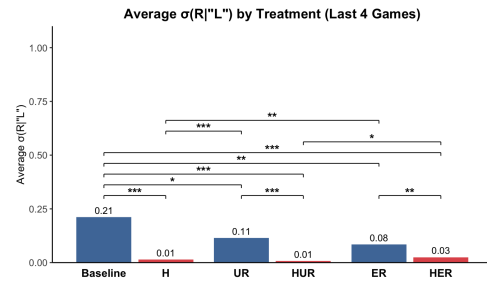


(d) Last 4 Games - R1

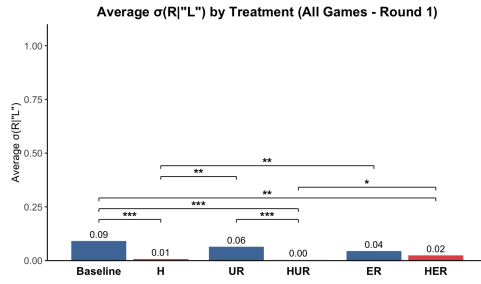
Figure A8. : Receiver Behavior: Action M given Message “ L ” (Session-level Ranksum).



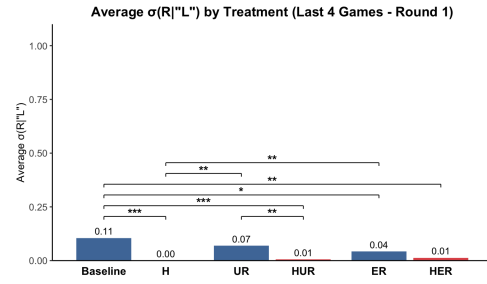
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

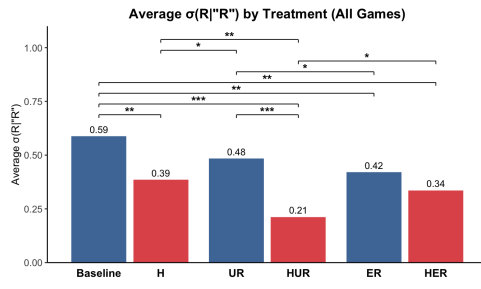


(c) All Games - R1

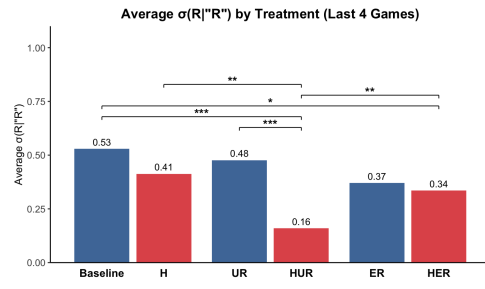


(d) Last 4 Games - R1

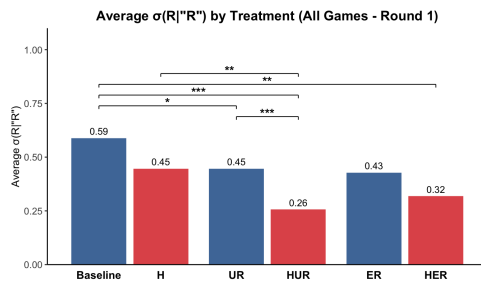
Figure A9. : Receiver Behavior: Action R given Message “ L ” (Session-level Ranksum).



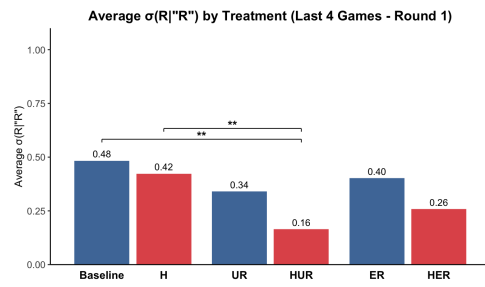
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds

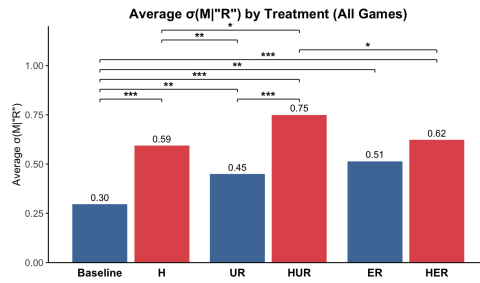


(c) All Games - R1

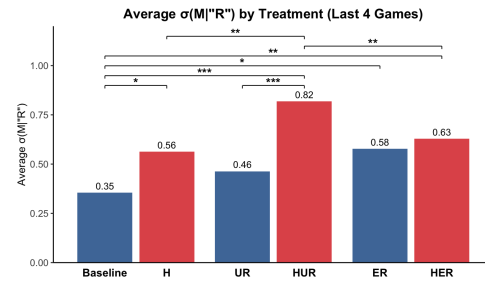


(d) Last 4 Games - R1

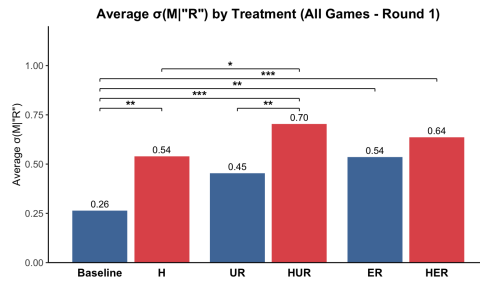
Figure A10. : Receiver Behavior: Action R given Message “ R ” (Session-level Ranksum).



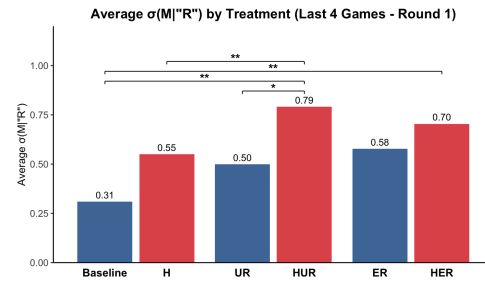
(a) All Games - All Rounds



(b) Last 4 Games - All Rounds



(c) All Games - R1



(d) Last 4 Games - R1

Figure A11. : Receiver Behavior: Action M given Message “ R ” (Session-level Ranksum).

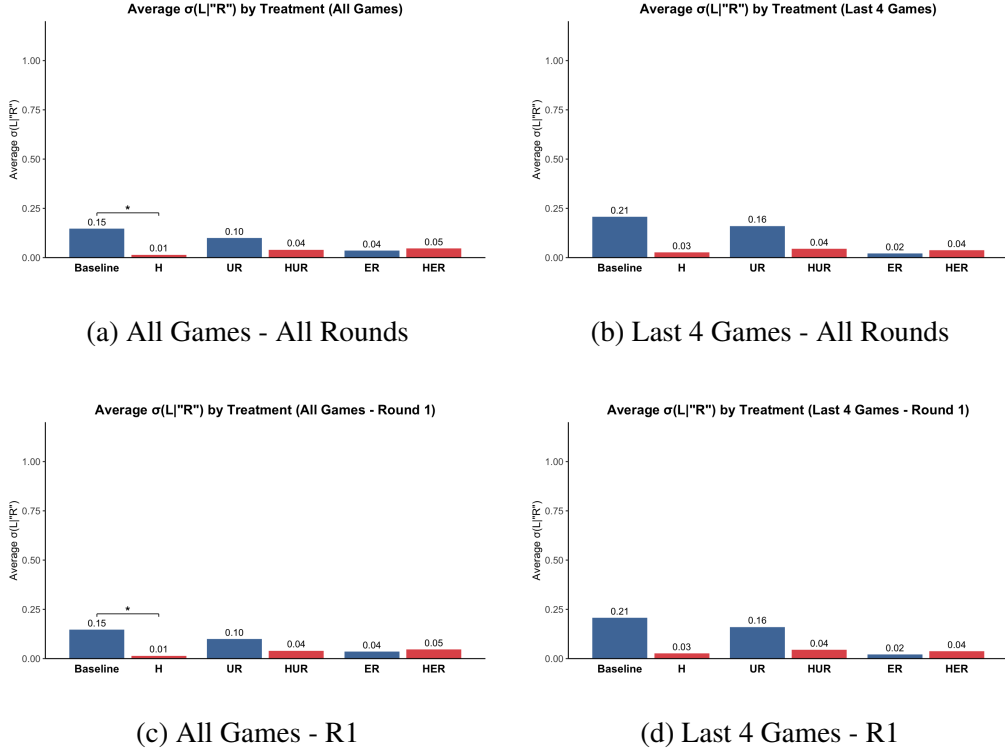


Figure A12. : Receiver Behavior: Action L given Message “ R ” (Session-level Ranksum).

A5. Derivation of the Minimal Required Time Discount Factor $\hat{\delta}$ for Blind Trust Equilibrium

We consider a generalized version of the Blind Trust Equilibrium (BTE), whose on-path strategy is: *always sending message “ L ” as Sender*, and *always following message is “ L ”*. Since Sender always has no incentive to deviate to BE, we only need to analyze two possible Receiver one-shot deviations to BE.

CASE 1. — Player A (as the current Receiver) deviates by choosing action M rather than L in response to message “ L ”.

- No deviation:

$$0.5 + \frac{\delta}{1-\delta} (\gamma + 0.5(1-\gamma))$$

- Deviation:

$$b + \frac{\delta}{1-\delta} (\gamma(1-b) + (1-\gamma)b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2b-1}{(4b-1)\gamma}$$

CASE 2. — Player B (as the current Receiver) deviates by choosing action M rather than L in response to message “ L ”.

- No deviation:

$$0.5 + \frac{\delta}{1-\delta} ((1-\gamma) + 0.5\gamma)$$

- Deviation:

$$b + \frac{\delta}{1-\delta} ((1-\gamma)(1-b) + \gamma b)$$

- Incentive compatibility condition:

$$\delta \geq \frac{2b-1}{(4b-1)(1-\gamma)}$$

These two inequalities jointly determine the minimal discount factor $\hat{\delta}$ needed to sustain BTE. Note that given $\gamma \geq 0.5$ in our setting, the binding constraint is case 2. In ER and UR, with $b = 0.5$, the binding constraint is reduced to $\delta \geq 0$ which is trivial. In HER and HUR, with $b = 0.9$, the binding constraint is reduced to $\delta > 8/13 \simeq 0.62$ for HER and $\delta \geq 8/7.8 \simeq 1.03$ for HUR (which is never satisfied). Therefore, BTE is as robust as BE in ER and UR, while BTE is only supported in HER when $\delta > 0.62$ and is never supported in HUR.

A6. Trend across Rounds

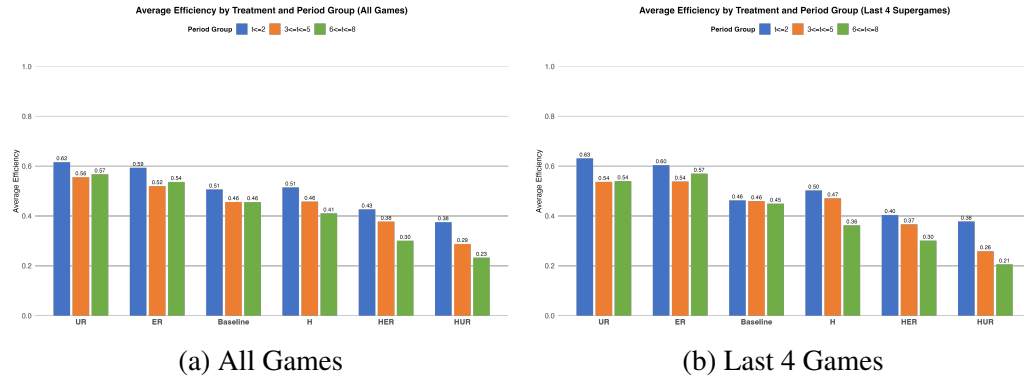
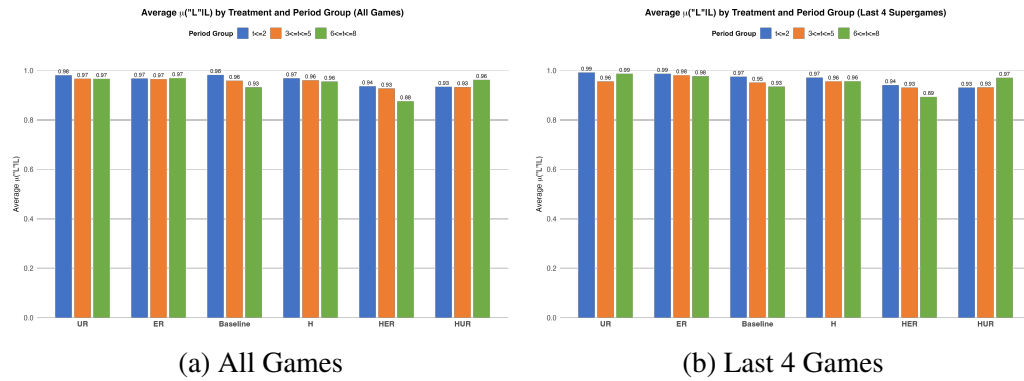
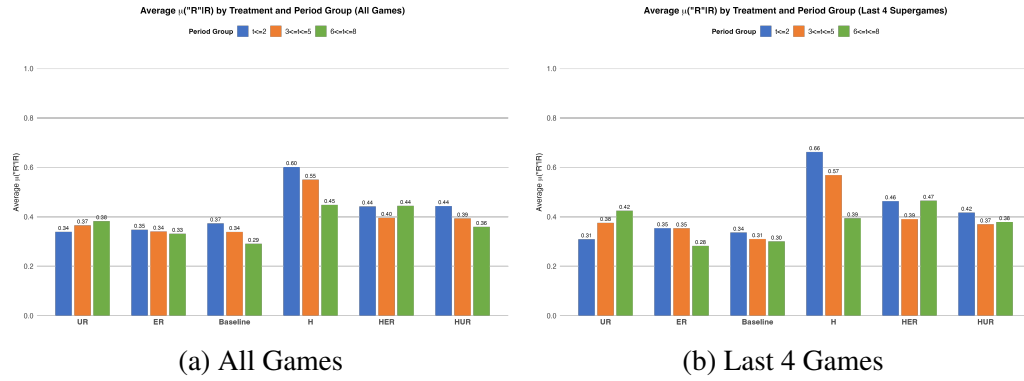
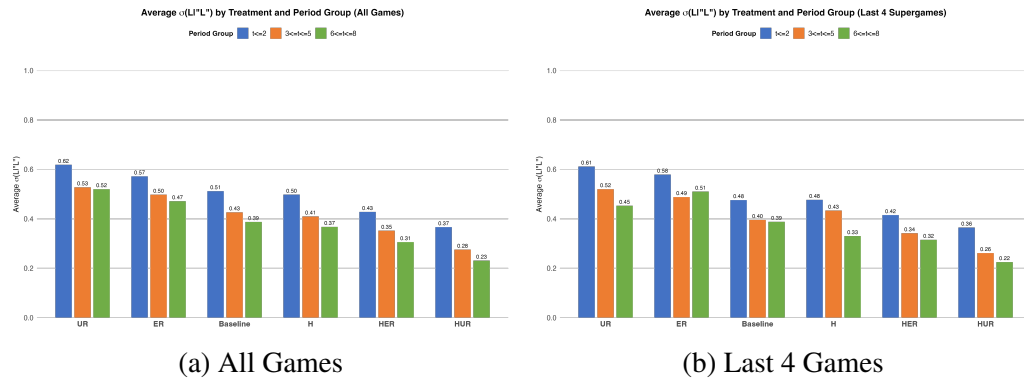
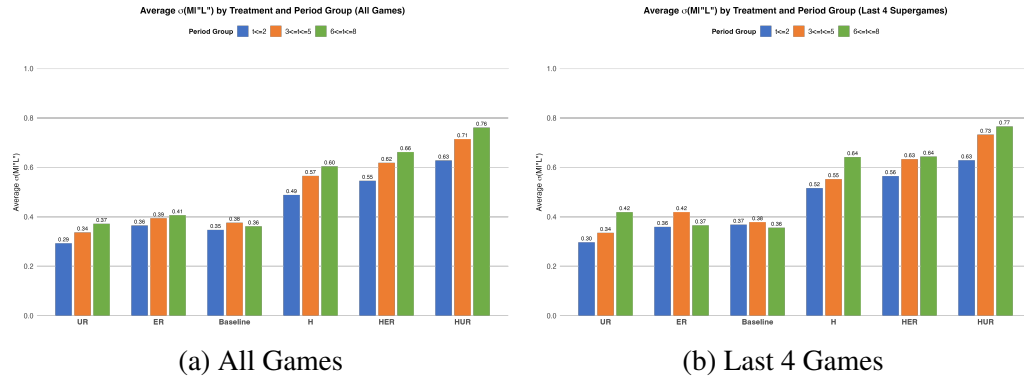
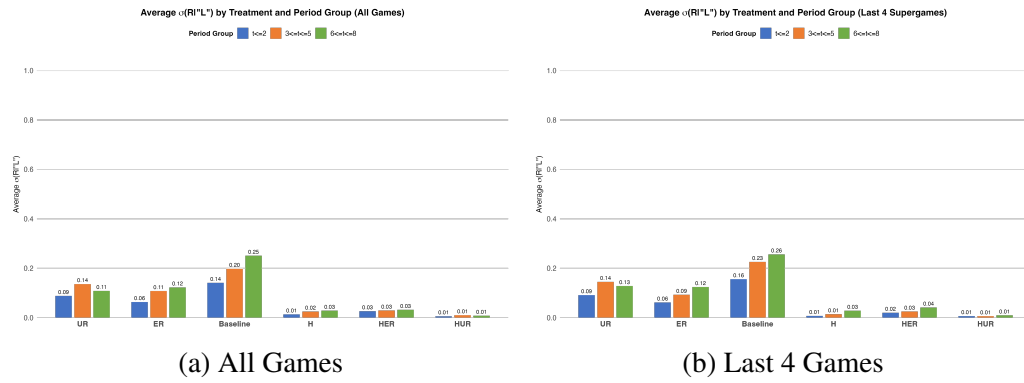
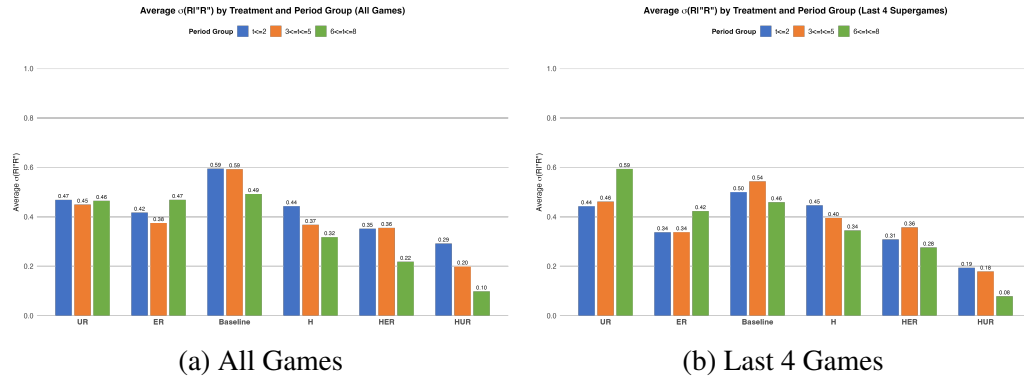
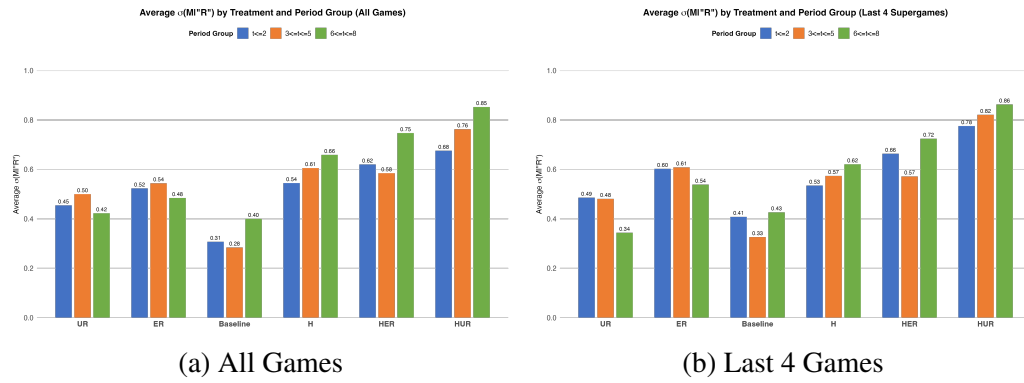


Figure A13. : Round trend: Efficiency

Figure A14. : Round trend: $\mu("L"|L)$

Figure A15. : Round trend: $\mu("R"|R)$ Figure A16. : Round trend: $\sigma(L|L)$

Figure A17. : Round trend: $\sigma(M|“L”)$ Figure A18. : Round trend: $\sigma(R|“L”)$

Figure A19. : Round trend: $\sigma(R|R)$ Figure A20. : Round trend: $\sigma(M|R)$

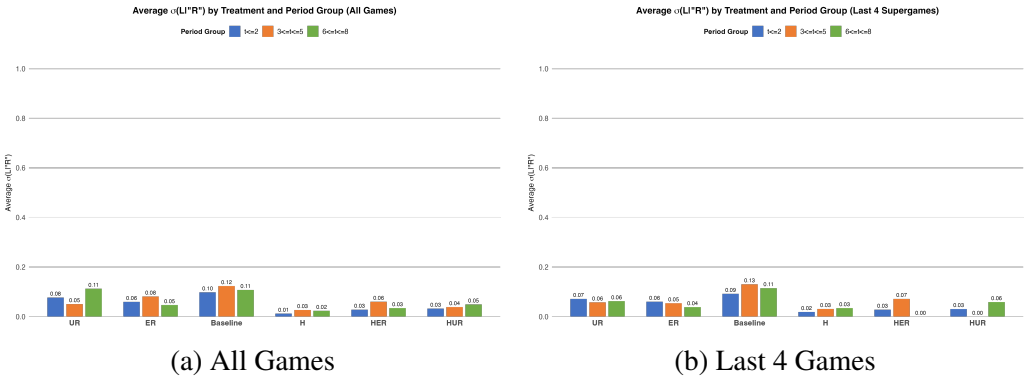


Figure A21. : Round trend: $\sigma(L|R)$